

TVS Group (TSF Lineage) Consolidated AI Immersion Knowledge Pack

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TVS Group (TSF Lineage) — Consolidated AI Immersion Knowledge Pack

TCS “Rapid Build” AI Immersion · Master Reference (Expanded Edition)

This document consolidates seven parallel deep-research workstreams into a single, non-overlapping reference and adds the nine-table participant clustering, the 27-card challenge portfolio, and a build-readiness assessment. It is decision-support for facilitators and the leadership room — not a strategy recommendation, investment thesis, or solution design.

Scope lineage: the **T.S. Santhanam branch** of the erstwhile TVS family (“TSF Group”) — Sundaram Finance, Brakes India, T.V. Sundram Iyengar & Sons and their subsidiaries. The TVS Motor / Sundaram-Clayton (Srinivasan), Sundram Fasteners (Krishna), and TVS Mobility / Srichakra (Rajam) lineages are out of scope.

Conventions: Indian/British English; INR in crore/lakh as published. Recency window January 2024 – June 2026. Confidence tags used throughout — **Verified** (primary/regulator/exchange source, or two independent sources), **Inferred** (reasoned from indirect evidence), **Assumed** (hypothesis to confirm live at the immersion).

Provenance of this pack

Workstream	Coverage	Research engine
WS0	Group topology, lineage & source strategy	Claude
WS1	Manufacturing & auto components	Gemini
WS2	Financial services — lending	ChatGPT
WS3	Investments & insurance	ChatGPT
WS4	Auto distribution & aftermarket	Claude
WS5	Shared services & social	Gemini
WS6	Group AI & digital posture	ChatGPT

Each fact appears once, in its natural home. The lineage and disambiguation rules (Part A) and the group AI posture and cross-entity constraints (Part B) are stated centrally and assumed everywhere after. Parts C1–C5 are domain deep-dives with no entity-fact repetition between them. Part D maps the room into nine tables; Part E is the challenge portfolio; Part F is the build-readiness assessment; Part G is the consolidated gap, reconciliation and confidence log.

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Executive Summary & How to Use This Pack

What this is. A single, consolidated ground-truth reference for the TCS “Rapid Build” AI Immersion with the **T.S. Santhanam branch of the TVS family (“TSF Group”)** — the lineage anchored on Sundaram Finance and T.V. Sundram Iyengar & Sons. It merges seven parallel deep-research workstreams, the nine-table participant clustering, the 27-card challenge portfolio and an independent build-readiness assessment into one non-overlapping document.

The five things to hold in mind.

1. **Scope is defined by family branch, not brand.** The in-scope cluster is the Santhanam branch (Sundaram Finance, Brakes India, Wheels India, IMPAL, Turbo Energy, Axles India, the Sundaram financial-services subsidiaries, and the social arms). TVS Motor / Sundaram-Clayton (Srinivasan), Sundram Fasteners (Krishna) and TVS Mobility / myTVS / Sundaram Honda (Rajam) are **out of scope** despite shared “TVS”/“Sundaram” branding. Resolve every fact to a named legal entity and CIN.
2. **The posture is federated.** There is no group-level AI strategy or AI centre of excellence; AI is owned entity by entity. The clearest disclosed AI ambition sits in **lending and insurance**; the clearest industrial-digital substrate sits in **manufacturing**. Two named anchor mandates carry the agenda: the **Sundaram Finance CTDO** (board-connected, data-lake/ML/agentive pilots) and the **Brakes India CIDO** (manufacturing digitisation).
3. **These are not blank canvases.** Sundaram Finance already runs a cloud data lake, ML models and agentive pilots; Royal Sundaram already uses ML and RPA in claims and has modernised its core on private cloud; Turbo Energy already runs edge-AI adaptive CNC control with measured savings. Any idea that re-proposes these would duplicate live work — the do-not-duplicate scans in Parts B and C matter.
4. **Regulation is a hard design constraint.** RBI Digital Lending Directions (India-only data, 24-hour foreign-data deletion), IRDAI cyber rules (6-hour incident reporting), SEBI CSCRf, and the DPDP Act 2023 + Rules (Nov 2025) bound any build that touches lending, insurance, mutual-fund or personal data. Keep human-in-the-loop on regulated decisions.
5. **The build-ready slices are known.** Of the 27 challenges, the safest in-room primaries are casting-defect traceability (T1-S1), turbo-unbalance root-cause (T3-P1), borrower-document interpretation (T5-P1), field-collection targeting (T6-P1), claims orchestration (T7-P1) and shared-services automation (T9-S2). The single most group-unlocking — and hardest — idea is cross-entity master-data/entity resolution (T9-P1).

How to navigate. Part A is the entity/lineage ground truth (read once); Part B is the group AI posture and the regulatory guardrails (read once); Parts C1–C5 are the five domain deep-dives; Part D maps the 78 leaders into nine tables; Part E is the challenge portfolio; Part F is the build-readiness assessment; Parts G–L are the gap log, leadership directory, function-by-function implications, facilitator notes, source register and inference log.

Group at a Glance

A consolidated scale snapshot of the in-scope entities (latest disclosed figures; see the relevant domain part for context and confidence).

Entity	Domain	Headline scale (latest disclosed)
Sundaram Finance	Lending (NBFC)	600 branches; FY26 deposits ₹8,873 cr; net stage-3 0.69%; CV 46% / cars 26% / CE 11% of book; 4,791 employees
Sundaram Home Finance	Housing finance (HFC)	115 branches (EB 72); FY26 AUM ₹19,909 cr; disbursements ₹6,842 cr; gross stage-III 1.11%; 1,838 employees
Sundaram Asset Management	Mutual fund	₹76,008 cr avg AUM (₹85,000 cr+ site); 3.5M+ investors; 80 branches; 71,816 distributors
Sundaram Alternate Assets	PMS / AIF	~₹7,000 cr+ AUM; AIF ₹3,284 cr (4 Cat III + 5 Cat II); ~3,000 clients
Royal Sundaram	General insurance	FY25 GDP ₹3,763 cr; 27.39 lakh policies; 3.68 lakh claims; SF 50% / Ageas 40%
Brakes India	Braking & foundry	FY25 ~₹7,509 cr; 16 plants; ~13% op. margin; foundry yield 55-65%
Wheels India	Wheels & structures	FY26 standalone ~₹5,098 cr (+15%); 4 plants + WIL USA; debt/EBITDA ~3.4x
Turbo Energy	Turbochargers	FY25 op. income ~₹2,960 cr; ~67% domestic share; ~9.3% net margin; ₹986 cr free cash
Axles India	Axles	FY25 ~₹841.5 cr; 62.98% TSF (Dana exit 2025); 400k housings/yr
ABI-Showatech	Castings / aerospace	FY25 op. income ~₹554.8 cr; aerospace pivot; capex ₹80-100 cr
IMPAL	Parts distribution	FY25 sales ₹788.81 cr (+6.4%); profit ₹83.67 cr (+13.2%); ~89 branches; ~18,000 dealers
Sundaram Business Services	BPM / shared services	400+ professionals; SMSF ~5,000 funds/yr; mortgage 3,600+ brokers; NASSCOM Expert (3.10)
Sundaram Medical Foundation	Healthcare (social)	230+ beds; 30+ specialties; DNB teaching; 30% free OP / 15% free IP; NASSCOM Enthusiast (2.50)
Laxmi Charities	Education (social)	200,000+ students over 5 decades; ~two-thirds female; NASSCOM Explorer (1.80)

Group identity (as publicly stated): revenue cited above ₹29,000 crore, 42,000+ employees, 1,200+ branches, 36+ factories — though figures vary across public pages, itself a tell of federated metadata ownership.

PART A — Group Topology & Disambiguation Ground Truth

A.1 The one finding that governs everything

The in-scope cluster is the **T.S. Santhanam branch** of the TVS founding family — today self-styled the **“TSF Group.”** Its two control pillars are **T.V. Sundram Iyengar & Sons Private Limited (TVS & Sons)** on the auto/distribution side and **Sundaram Finance Limited** on the financial-services side, with **Sundaram Finance Holdings Limited (SFHL)** and **Trichur Sundaram Santhanam & Family Private Limited (TSSFPL)** as the apex holding vehicles consolidated through the 2020–2022 family settlement. **Every “Sundaram-Clayton” or “TVS Motor” reference belongs to the T.S. Srinivasan branch and is OUT of scope.**

The roster: **20 confirmed in-scope corporate/charitable entities** plus two unincorporated distribution divisions (Sundaram Motors, Madras Auto Service) and the apex holding company. Of the corporate entities, **four are listed on equity exchanges** — Sundaram Finance and Sundaram Finance Holdings (financial side), Wheels India and IMPAL (auto/distribution side); the remainder are unlisted/private, with Sundaram Home Finance an unlisted-equity but listed-NCD issuer. Regulator spread: most auto-components entities answer only to MCA/ROC; the financial-services cluster spans RBI (NBFC), RBI/NHB (HFC), SEBI (AMC and PMS/AIF) and IRDAI (general insurance).

A.2 The founder and the four branches

The group traces to **Thirukkurungudi Vengaram Sundram Iyengar (1877–1955)**, who founded **T.V. Sundram Iyengar & Sons in 1911** (the group’s first passenger bus service launched 1912, Madurai–Pudukottai). His four surviving sons — **T.S. Rajam, T.S. Krishna, T.S. Srinivasan and T.S. Santhanam** — became heads of four enduring branches (a fifth son, T.S. Doraisamy, died young). Over six decades a complex ownership web formed, mediated through three original promoter companies: TVS & Sons, Southern Roadways and Sundaram Industries.

Founding son / family	Branch businesses (illustrative)	Scope
T.S. Santhanam	Sundaram Finance, Sundaram Finance Holdings, Brakes India, Wheels India, IMPAL, Sundaram Motors, Madras Auto Service	IN
T.S. Srinivasan (Venu Srinivasan)	Sundaram-Clayton, TVS Motor Company, TVS Electronics (apex: TVS Holdings Ltd)	OUT
T.S. Krishna (Suresh Krishna)	Sundram Fasteners and allied	OUT
T.S. Rajam	TVS Srichakra, TVS Logistics → TVS Mobility (incl. Sundaram Honda, myTVS)	OUT
K. Mahesh family	Sundaram Brake Linings (via Madurai Alagar Enterprises)	OUT
K. Ramesh family	Southern Roadways	OUT

A.3 Where and when the split occurred

The historic “live separately, yet work together” understanding was formalised through a **Memorandum of Family Arrangement signed in 2020** (approved by the NCLT on 4 February 2022 per WS4’s reading), with corporate restructuring (demergers and amalgamations) finalised through 2022. Two foundational agreements were executed on **29 January 2021**: a brand agreement governing use of the “TVS,” “Sundaram” and “Sundram” trademarks (no royalty among operating companies), and a Non-Competition Agreement among the family groups. A key milestone was the Sundaram-Clayton restructuring scheme approved at a board meeting on **9 February 2022**.

Under the settlement, **TSSFPL** took ownership of **Wheels India, Brakes India, IMPAL, Sundaram Motors, Madras Auto Service and the automotive business of TVS & Sons** — the in-scope auto-components-and-distribution cluster. The same Santhanam branch controls **Sundaram Finance** and **Sundaram Finance Holdings**. Because the “TVS”/“Sundaram” brands can be used by all branches, brand overlap is precisely why identity-confusion risk is high and why **entity-level (not brand-level) attribution is mandatory**.

A.4 The four mandatory disambiguation rules

- “Sundaram Finance”** = in-scope apex financial-services NBFC (NSE: SUNDARMFN; CIN L65191TN1954PLC002429), Santhanam branch. Never merge with any “Sundaram-Clayton” company.
- “Sundaram-Clayton”** = OUT (Srinivasan / TVS Motor branch). The historic listed Sundaram-Clayton Ltd was **renamed TVS Holdings Limited effective June 2023** (BSE scrip 520056 from 24 August 2023); a separate new company carries the die-casting business under the “Sundaram-Clayton Limited” name.
- “Sundaram Motors”** = in-scope vehicle-dealership **division of TVS & Sons** (est. 13 August 1945; the **Mercedes-Benz** franchise in Tamil Nadu and Karnataka). NOT TVS Motor Company. The **“Sundaram Honda”** network belongs to the Rajam branch’s TVS Mobility and is OUT despite the shared name.
- “TVS”** as a bare token is ambiguous: resolve only via a named legal entity + CIN; reject and quarantine any generic “TVS Group” source that blends lineages.

A.5 Full in-scope entity roster

All confirmed CINs registered at ROC-Chennai. Clusters: AC = auto components, FS = financial services, DIST = distribution, PHIL = philanthropy, HOLD = holding/apex.

#	Legal name	CIN	Incorp.	Registered office	Cluster	Listing / basis
1	T.V. Sundram Iyengar & Sons Pvt Ltd	U34101TN1929PTC002973	1929 (month to confirm)	TVS Building, West Veli St, Madurai	HOLD/DIST	Unlisted; apex promoter co
2	Sundaram Finance Ltd	L65191TN1954PLC002429	1954	21 Patullos Road, Chennai	FS	Listed BSE 590071 / NSE SUNDARMFN; ISIN INE660A01013
3	Sundaram Finance	L65191TN1954PLC002429	1954	21 Patullos Road	FS/HOLD	Listed NSE

3	Sundaram Finance Holdings Ltd	L651001TN1993PLC025998	1993	21 Patullos Road, Chennai	FS/HOLD	Listed NSE SUNDARMHLD; ISIN INE202Z01029
4	Brakes India Pvt Ltd	(to confirm)	1962	Padi, Chennai	AC	Unlisted; core JV
5	Wheels India Ltd	(equity CIN to confirm; ISIN INE715A01015)	1960	21 Patullos Road, Chennai	AC	Listed BSE 590073 / NSE WHEELS
6	Turbo Energy Pvt Ltd	U40107TN1982PTC009363	1982	67 Chamiers Road, Chennai	AC	Unlisted; JV BIPL+SF+BorgWarner
7	Axles India Ltd	U27209TN1981PLC008630	1981	21 Patullos Road, Chennai	AC	Unlisted (public co)
8	ABI-Showatech (India) Pvt Ltd	U40103TN1991PTC021872	1991	67 Chamiers Road, Chennai	AC	Unlisted
9	Sundaram Composite Structures Pvt Ltd (Leggero)	U36100TN2010PTC076378	2010	Padi, Chennai	AC	Unlisted; carbon-fibre JV
10	BIADS India Pvt Ltd	U29301TN2024PTC171099	2024	C/o Brakes India, Padi	AC	Unlisted; Brakes India + ADVICS JV
11	Sundaram Dynacast Pvt Ltd	U28920TN1993PTC025794	1993	Ambattur, Chennai	AC	Unlisted; JV with Form Technologies
12	India Motor Parts & Accessories Ltd (IMPAL)	L65991TN1954PLC000958	1954	46 Whites Road, Chennai	DIST	Listed NSE IMPAL / BSE 590065
13	Sundaram Home Finance Ltd	U65922TN1999PLC042759	1999	21 Patullos Road, Chennai	FS	Unlisted equity; listed NCDs
14	Sundaram Asset Management Co Ltd	U93090TN1996PLC034615	1996	21 Patullos Road, Chennai	FS	Unlisted; SEBI AMC
15	Sundaram Alternate Assets Ltd	(to confirm)	(to confirm)	Chennai	FS	Unlisted; SEBI PMS/AIF
16	Royal Sundaram General Insurance Co Ltd	U67200TN2000PLC045611	2000	21 Patullos Road, Chennai	FS	Unlisted; SF 50% / Ageas 40% / SF-grp 10%
17	Sundaram Business Services Ltd	U74140TN2005PLC057179	2005	21 Patullos Road, Chennai	FS/SVC	Unlisted; SFHL 100%
18	Sundaram Motors (division of TVS & Sons)	n/a (division)	1945	C/o TVS & Sons	DIST	Unincorporated division
19	Madras Auto Service (division of TVS & Sons)	n/a (division)	1936	C/o TVS & Sons	DIST	Unincorporated division
20	Sundaram Medical Foundation	Public Charitable Trust (Deed 29 Nov 1990)	1990	Anna Nagar, Chennai	PHIL	Trust
21	Laxmi Charities	Charitable trust	1973	Chennai	PHIL	Trust

The “21 Patullos Road, Chennai 600002” address recurs across the financial-services entities and several auto holdings — the Sundaram Finance group’s registered seat, and a useful (not sufficient) disambiguation cue versus the historic Sundaram-Clayton/TVS Motor address. “67 Chamiers Road” links Turbo Energy and ABI-Showatech (common turbocharger lineage). One genuine open candidate — **Flometallic India Pvt Ltd** (a Brakes India co-shareholder alongside TVS & Sons, SFHL and IMPAL) — is flagged for live confirmation as a possible in-lineage addition.

A.6 Holding & cross-holding structure

Two apex poles, both Santhanam-controlled.

Pole A — Financial services, anchored on Sundaram Finance Ltd. No single apex holding vehicle controls Sundaram Finance: per its SEBI Reg.31(4) disclosure to NSE dated 2 April 2026, total promoter and promoter-group shareholding is **4,13,38,283 shares = 37.21%**, split between ~84 family individuals/HUFs (~23.8%) and ~10 small family holding companies (~12.7%) — the largest corporate holders being IMPAL, Raghuvamsa Holdings, Uthirattadhi Sriram Holdings, Silver Oak Holdings, Rohini Holdings, Padmalakshmi Holdings, Allegro Holdings, Revathi Holdings, Azorious Holdings and Maham Holdings. (Note: “Sundaram Finance Investments Ltd” is the defunct interim 2016 name of Sundaram Finance Holdings Ltd, renamed March 2017 — not a separate live entity.)

Sundaram Finance Ltd directly controls the regulated financial subsidiaries: **Sundaram Home Finance** (100%; BNP Paribas Personal Finance’s 49.9% bought back 30 Sep 2019), **Sundaram Asset Management** (100%) → **Sundaram Alternate Assets** (100%, PMS/AIF), and **Royal Sundaram** (50%; **Ageas Insurance International N.V., Belgium, 40%** acquired Feb 2019; other SF-group corporates 10%).

Sundaram Finance Holdings (SFHL) is the auto-components investment arm — Wheels India (~23.85%, FY23), Axles India (raised to **62.98% in 2025**), Turbo Energy — and the **100% parent of Sundaram Business Services**.

Pole B — Auto components and distribution, anchored on TVS & Sons / TSSFPL. TSSFPL owns Wheels India, Brakes India, IMPAL, Sundaram Motors, Madras Auto Service and the automotive business of TVS & Sons. **Wheels India** is jointly promoted by TSSFPL (~30.31%, FY23) and SFHL (~23.85%). **Brakes India** (after ZF Friedrichshafen’s exit, completed 22 June 2021) is held by TVS & Sons, SFHL, IMPAL and Flometallic India, and in turn owns/co-owns Turbo Energy, Sundaram Dynacast, Sundaram Composite Structures and the 2024 BIADS JV.

A.7 Listed-status & regulator map

Entity	Listing	Regulator(s)	Key registration
Sundaram Finance	Equity (BSE+NSE)	RBI (NBFC, deposit-taking) + SEBI	NBFC-Middle Layer; CoR to confirm

Sundaram Finance Holdings	Equity (NSE)	MCA + SEBI; RBI CIC status to confirm	—
Wheels India	Equity (BSE+NSE)	MCA + SEBI	—
IMPAL	Equity (NSE)	MCA + SEBI	—
Sundaram Home Finance	Listed NCDs	RBI (HFC) / NHB supervision	CoR under s.29A NHB Act; HFC-ML
Sundaram Asset Management	Unlisted	SEBI (Mutual Fund)	Reg. MF/034/97/2
Sundaram Alternate Assets	Unlisted	SEBI (PMS + AIF)	PMS INP000006271; AIF Cat II & III trusts
Royal Sundaram	Unlisted	IRDAI	Reg. No. 102 (23 Oct 2000) — first private general insurer
Auto-components + TVS & Sons	Mostly unlisted	MCA/ROC only (+ SEBI for listed)	CINs per roster
Sundaram Medical Foundation / Laxmi Charities	Trusts	Charitable-trust framework	no MCA CIN

A.8 Source-tier strategy & confidence ceilings

- **Listed equities** (Sundaram Finance, SFHL, Wheels India, IMPAL): Tier-1/2 — SEBI/exchange disclosures and audited annual reports; the highest-confidence entities.
- **Listed-debt issuer** (Sundaram Home Finance): audited ARs under SEBI LODR + NHB HFC list.
- **SEBI-regulated FS subsidiaries** (Sundaram AMC, Sundaram Alternate Assets): SEBI registers for licence status; issuer ARs for parentage/AUM (100% intragroup ownership).
- **IRDAI insurer** (Royal Sundaram): IRDAI register + registration number on policy documents.
- **Unlisted/private auto-components**: Tier-1 for identity (MCA CIN/registered office/directors); **lower confidence on ownership percentages**, which are not public.
- **Apex private holding** (TVS & Sons, TSSFPL): Tier-1 for identity; ownership opaque (press-sourced family allocation, to confirm).
- **Philanthropy**: court/issuer material for legal form; no MCA CIN exists.

A.9 Risk register of likely identity confusions (ranked)

1. **Sundaram Finance (IN) vs Sundaram-Clayton (OUT)** — both “Sundaram-”; both “TVS.” Confirm by CIN.
2. **“Sundaram Motors” (IN division) vs TVS Motor Company (OUT)** — dealership division (1945) vs listed two-wheeler maker.
3. **Sundaram Finance Holdings (IN) vs defunct “Sundaram Finance Investments Ltd”** — same entity, renamed 2017; not two live entities.
4. **TVS & Sons shared board** — directors span all branches; director overlap ≠ control. Attribute via the 2020–2022 settlement.
5. **Sundram Fasteners / Sundaram Brake Linings (OUT)** — Krishna and Mahesh families respectively.
6. **“Sundaram Motors” Mercedes (IN) vs “Sundaram Honda” (OUT, TVS Mobility/Rajam)** — same brand token, different branch.

PART B — Group AI & Digital Posture

B.1 The posture is federated, not unified

There is **no public group-level AI strategy** at the TSF / TVS & Sons layer. TSF Group’s own site positions the lineage as a century-old house spanning automotive and financial services (“common values, diverse ventures”), with revenue cited variously above ₹29,000 crore (and an older page at ₹21,000 crore), **42,000+ employees, 1,200+ branches and 36+ factories** — a group identity statement, not a group AI statement. Even basic group facts are not uniformly maintained across public channels (Sundaram Finance’s own pages cite both “600 branches” and “nearly 1,000 branches”), a tell of decentralised metadata ownership.

AI posture is **federated by entity** and asymmetric. The single most useful framing for the room:

AI momentum is anchored in lending and insurance; build tractability may be highest in manufacturing. (Verified/Inferred, Medium-High confidence.)

Manufacturing shows strong industrial-digital depth but less explicit AI articulation, and is less encumbered by RBI/IRDAI/SEBI data constraints — so it may be the easier surface for a thin-slice build even though the clearest *AI ambition* is disclosed on the finance side.

B.2 Entity-by-entity posture

Sundaram Finance — the clearest technology/AI discloser in scope. Its annual report calls IT a “critical pillar,” and references a **400-plus-server data centre, ISO/IEC 27001:2022 accreditation, a new core-system foundation, selective cloud migration, intelligent document processing, a cloud-based data lake as the foundation of a “Systems of Intelligence” layer, more ML models, pilots using an agentic framework, a 24×7 SOC, a CISO, and board-level IT strategy oversight**. Strategy is explicitly “high touch and high tech,” not pure automation.

Royal Sundaram — the second-strongest discloser; an **agile, technology-first insurer**. Early AI and digitisation pilots “with promising results”; **ML and RPA** in operations and claims; omni-channel claims via bots/WhatsApp/SMS/email/app/web; **FirstGen v7 and NIP v3 with FirstApex on private cloud** (Sep 2025). Aug 2025 statements: the next chapter will use **AI and advanced analytics for personalised protection** and deepen phygital experiences — more explicit AI language than any manufacturing-side entity.

Brakes India — technology-led, operations-heavy manufacturer: ₹7,000 crore revenue, 19 facilities, 250+ engineering team, **zero information-security non-compliance FY23-24**. Disclosed estate: **AMTIX** (in-house machine-building / configurable automation), robotics for machining/assembly/test, vision systems, **QR-code verification and cloud-server-based traceability**, SAP S/4HANA migration, Siemens Teamcenter. What is missing is an explicit enterprise-AI programme statement.

Sundaram Asset Management / Sundaram Mutual — strong digital distribution/servicing (consumer **Investor App**; **InvWise** distributor platform with real-time AUM insights, NFO/SIP dashboards, enhanced reports, digital mandates, non-financial transactions), but **no AI strategy comparable to SF or Royal Sundaram** in surfaced material. Public signal = channel maturity, not AI ambition.

Distribution entities (IMPAL, Sundaram Motors) — functioning digital surfaces and service interfaces (online test-drive/careers forms; B2B dealer ordering at IMPAL), but no reusable AI estate or named AI programme.

Shared & social — weakest public evidence of a group AI/data platform or central AI office.

B.3 Disclosed AI/digital estate (no group CoE evidenced)

Entity / asset	What exists publicly	Status
Sundaram Finance	Cloud data lake, ML models, agentic pilots, IDP, reusable workflows, task management, communication framework, nano-learning, new core-system foundation, 400+ server DC, SOC, CISO, selective cloud	Scaling — most mature disclosed AI-adjacent estate in scope
Royal Sundaram	AI/digitisation pilots, ML + RPA in ops/claims, omni-channel claims bots, FirstGen v7 + NIP v3 on private cloud	Scaling — technology-forward insurer
Brakes India AMTIX	In-house machine-building, configurable automation, robotics, vision, cloud traceability, QR verification	Mature for automation; experimental for AI
Sundaram Mutual platforms	Investor App, InvWise distributor platform, dashboards, digital mandates	Mature for digital distribution; AI story weak
Group-wide AI platform / CoE	No public disclosure found	Not claimed — federated ownership

B.4 The two anchor mandates

- Sundaram Finance — Chief Technology & Digital Officer, Sudheer Warriier** (joined 03-01-2022; ~30 years’ IT including TCS). Member of the **IT Strategy Committee**; delivered a board presentation on IT and cyber initiatives on **27 March 2025**. Owns a broad technology and digital-transformation estate with board oversight, cyber accountability and explicit AI-adjacent initiatives. Supported by a Head – Technology Shared Services, a GM & Head – Digital Initiatives, an analytics leader, and Sundaram Infotech Solutions managers. **Verified, High.**
- Brakes India — SVP & Chief Information & Digital Officer, Ramya Sampathkumar** (appointed September 2024; ~20 years across Cognizant and GMMCO). Elevated onto the public leadership page alongside business-unit heads — a serious organisational tell that information/digital capability is now treated as strategic. Public evidence is thinner on remit/reporting line; the role likely bridges enterprise IT with manufacturing digitisation/OT, but that is an inference. Primary mandate (per WS1): define a unified industrial data schema across the group’s SAP S/4HANA migrations and champion a shared platform for scalable AI to prevent duplicative builds. **Verified title / Inferred remit, Medium.**

Taken together the mandates signal a **federated operating model** with strong entity-level leadership and no group AI office. The room’s narrative should avoid claiming parity of AI maturity between the two: *finance has the clearest public AI story; manufacturing has the clearest public industrial-digital substrate.*

B.5 Cross-entity hard constraints (the guardrails every AI build inherits)

- **RBI Digital Lending Directions, 2025** (Sundaram Finance / Sundaram Home): lending service providers must not store borrower data beyond minimal basics; the RE remains responsible for privacy/security; **all data stored only on servers in India**; any data processed outside India deleted from foreign servers and repatriated within **24 hours**; biometric-storage limits; credit reporting required for digital lending.
- **RBI IT Governance, Risk, Controls & Assurance Directions, 2023**: board-governed IT strategy/risk/control for NBFCs. RBI IT-outsourcing and 2025 NBFC-outsourcing directions impose vendor accountability.
- **IRDAI Information & Cyber Security framework** (Royal Sundaram): board-approved cyber policy; periodic testing/audit (CERT-In empanelled firm); cyber incidents reported to IRDAI **within 6 hours**; logs retained **180 days**; board reporting. The 2024 Information Products and Policyholder-Protection master circulars and the 2026 Information & Cyber Security Guidelines apply to any AI touching claims/underwriting/customer records.
- **SEBI Cybersecurity & Cyber Resilience Framework (CSCRF), Aug 2024** (Sundaram AMC): consolidated, strengthened cyber expectations on AMCs with glide-path adoption through 2025; plus SEBI MF / AIF / PMS master circulars.
- **DPDP Act 2023 + Rules (notified 14 Nov 2025)**: the common data-privacy thread. Free/specific/informed consent with clear notice; purpose limitation and data minimisation; erasure on purpose-fulfilment or consent withdrawal; reasonable security safeguards; breach notification to the Data Protection Board of India; penalties up to **₹250 crore** (failure to prevent breach) and **₹200 crore** (failure to notify / children’s-data violations). A **child = under 18**; processing children’s data requires verified parental consent and prohibits tracking/behavioural-monitoring/targeted advertising — acute for the social arms and any cross-entity data sharing.

B.6 Responsible-AI maturity gap

Public governance is **cyber/privacy/operational-risk led, not model-risk led**. No in-scope entity publicly discloses a responsible-AI policy, model inventory, bias-testing framework, AI-specific audit methodology, model-risk committee, or board charter covering AI fairness/explainability/third-party model governance. The closest posture is “cyber/privacy-first governance with some AI experimentation.” For group-grade AI deployment, governance will need to move **beyond information security into model governance and cross-entity decision rights**. (Inferred from absence of public disclosure, not proof such controls are absent privately.)

B.7 Digital & AI milestones (group)

Period	Milestone
Apr 2024	Sundaram Finance IT Strategy Committee disclosed with Sudheer Warriar as member
FY 2024-25	Sundaram Finance discloses cloud data lake, more ML models, agentic pilots
Mar 2025	Warriar presents IT/cyber initiatives to SF independent directors
FY 2024-25	Royal Sundaram reports early AI/digitisation pilots, ML + RPA
Aug 2025	Royal Sundaram: next chapter to use AI/advanced analytics for personalised protection
Sep 2025	Royal Sundaram deploys FirstGen v7 + NIP v3 with FirstApex on private cloud
Sep 2024	Ramya Sampathkumar joins Brakes India as SVP & CIDO
2026 site state	Brakes India leadership page names CIDO; AMTIX/robotics/traceability visible

PART C1 — Manufacturing & Auto Components

C1.1 Cluster character and the EV transition

Eight operationally disciplined, safety-critical, JV-rich entities (TPM / Deming heritage): Brakes India, Wheels India, Turbo Energy, Axles India, ABI-Showatech, Sundaram Composites (Leggero), BIADS and Sundaram Dynacast. The Indian auto-component sector is in structural transition — tightening emissions, alternative-fuel powertrains, localised mechatronics — against moderating domestic OEM offtake and export headwinds. To defend margins in a lower-leverage environment, manufacturers pass raw-material increases (steel, pig iron, aluminium scrap) to OEMs with a contractual ~one-quarter lag, so performance hinges on reducing conversion cost, optimising energy and driving OEE.

The EV transition is both threat and catalyst. **Turbo Energy** still draws **86.4% of sales from diesel turbochargers** (down from 98.1% in FY20) and is pivoting to oil-free air compressors for hydrogen fuel cells and turbochargers for petrol/CNG/ethanol. **ABI-Showatech** hedges turbo decline by diversifying into precision **aerospace** (rotating, static and structural components). **BIADS** is the cluster’s explicit mechatronic-future vehicle (ESC/EPB for hybrid and BEV platforms).

C1.2 Per-entity profiles

Brakes India (BIPL). Fully TSF-owned; India’s largest braking-systems and safety-critical iron-castings maker. B2B across light vehicles, HCVs, off-highway/tractors and aftermarket. Processes: large-scale ferrous foundry (computerised green-sand moulding, DISA automatic-line pouring, automated core-making); high-precision CNC machining; mechatronic assembly; CED painting, zinc electroplating, electrocoagulation effluent treatment; dynamometer/ABS testing; in-house machine-builder **AMTIX**. Serves Maruti Suzuki, Hyundai, M&M, Tata Motors, Ashok Leyland, Volvo; **30%+ of European passenger vehicles use its castings**. Footprint: **16 plants** (Padi, Sholinghur, Naidupet, Polambakkam, Nanjangud) plus 51% foundry subsidiary Dunes Oman. **FY25 consolidated revenue ~₹7,509 crore; capex ₹500-600 crore/yr; operating margin moderated to ~13% (FY25) from 14.7% (FY24).**

Wheels India (WIL). Listed; one of India’s largest steel-wheel makers and a leading earthmover-wheel designer. Steel wheels, forged-aluminium wheels, pressed structures, bumpers, wind-turbine structurals. Processes: heavy structural forming/rolling, high-tonnage stamping, manual+robotic welding, CED/powder coating, forged-aluminium machining, fatigue/stress validation. Customers: Tata Motors, Ashok Leyland, tractor OEMs; off-highway exports to Caterpillar, Komatsu, John Deere. Plants at Padi, Sriperumbudur, Rampur, Jamshedpur; WIL USA Inc. (Delaware, 26 Jan 2024); Sundaram Hydraulics merged in (eff. 1 Sep 2023). **FY26 standalone revenue ~₹5,098 crore (+15%); capex ₹275-300 crore; debt/EBITDA ~3.4x (9M FY26).**

Turbo Energy (TEPL). 1982 JV of Brakes India, Sundaram Finance and BorgWarner; **~67% domestic turbocharger share**. Processes: 4-/5-axis CNC for compressor/turbine housings; rotor balancing up to 250,000 rpm; electron-beam/laser/plasma welding; single-piece-flow assembly with stage-level trace tracking. Customers: Maruti Suzuki (sole supplier in select diesel variants), M&M, Hyundai, Tata Motors, Ashok Leyland, BEML. Plants at Paiyanur, Pulivalam (TN) and Rudrapur (Uttarakhand). **FY25 operating income ~₹2,960 crore; ~9.3% net margin; ₹986 crore free cash/liquid investments.**

Axles India (AIL). 1981 JV (Eaton → Dana); **TSF Investments bought Dana’s stake in May 2025, raising holding to 62.98%** — now under complete TSF ownership. Heavy pressed/cast axle housings, lift axles, front beams for M&HCVs. Customers: Tata Motors, Ashok Leyland, Daimler India CV, Volvo; exports to Volvo/Dana. Plants at Sriperumbudur, Cheyyar and a 2025 greenfield at Jamshedpur. **400,000 axle-housings/yr; FY25 revenue ~₹841.5 crore (-1%).**

ABI-Showatech (ABI). 1991; specialised Tier-2 castings/machining (compressor/turbine wheels, brake cylinders) for global turbo/braking markets; aerospace diversification. Processes: aluminium gravity die-casting; stainless/superalloy investment casting; low-pressure and plaster-mould casting; 5-axis billet milling; NADCAP-approved HIP, FPI, X-ray, vacuum heat treatment. Supplies Turbo Energy and Brakes India internally; Tier-2 to BorgWarner globally. Integrated foundry+machining at Pulivalam, Ranipet. **FY25 operating income ~₹554.8 crore (from ₹869.3 crore) on a light-vehicle downturn; aerospace capex ₹80-100 crore/yr.**

Leggero (Sundaram Composite Structures). Nov-2020 JV with MIND S.r.l. (Bologna). Premium carbon-fibre structural parts for supercars, superbikes and medical imaging. Automated ply-cutting, cleanroom lay-up, autoclave curing, SMC compression moulding. Dual footprint: Bologna (design) + a 10,000 m² Chennai plant (Dec 2022).

BIADS. Incorporated 14 Jun 2024; Brakes India + ADVICS (Japan) JV for ESC, slip control and EPB. Cleanroom (Class 10,000) mechatronic assembly; high-speed PCBA testing; hydraulic-block machining; automated end-of-line functional-safety checks. Shoolagiri (Krishnagiri) plant inaugurated March 2026; authorised capital ₹300 crore, paid-up ₹275 crore; in ramp-up.

Sundaram Dynacast. Sep-1993 JV with Form Technologies (Dynacast). High-volume micro-precision metal components (automotive, medical, electronics, connectors) via proprietary multi-slide hot-chamber zinc die-casting, in-house micro-tooling and computerised finishing. Largest zinc HPDC facility in South India (68,000 sq ft, Ambattur). **FY24 revenue ₹150-200 crore; zero long-term external borrowings.**

C1.3 Value-chain operational pain map

Stage	Entities	Friction & metrics
Foundry & casting	BIPL, ABI	Casting yield only 55-65% (35-45% returns to remelt); melt-loss 2-4%/heat ; internal rejection 4-6% vs OEM sub-50-PPM targets; induction melting ~550-650 kWh/tonne (energy-cost exposed); ~2,00,000 t/yr foundry capacity
Machining	TEPL, AIL, ABI	Tool-wear trade-off: fixed-schedule swaps discard 20-30% tool life; run-to-failure scraps the workpiece; thermal/spindle drift breaches tolerances; long mixed-model changeovers degrade OEE
Assembly & test	TEPL, AIL, BIADS	Torque-verification errors on critical fasteners; dynamic unbalance found only at end-of-line; manual cycle-time variability starves downstream stations
Supply chain	WIL, AIL, BIPL	Static moving-average forecasting + high safety stock → bullwhip on abrupt OEM pull-rate swings; working capital trapped; raw-material volatility absorbed over a quarter

Maintenance & utilities	All	Reactive maintenance; single hydraulic-pump failure halts high-tonnage lines; lack of sub-metered energy monitoring prevents off-peak load shifting
Engineering / NPD	All	Sequential physical loop (tooling → prototype → machining → endurance) takes 12-18 months ; MD ambition to compress toward <10 months ; mechatronic skill gaps

C1.4 Data & systems estate (fragmented)

Entity	Core ERP	MES	SCADA/PLC	PLM/CAD	Primary gap
Brakes India	SAP S/4HANA (migrating)	Local MES	Centralised DISA foundry SCADA	Siemens Teamcenter	SCADA↔ERP gap; paper audits remain
Wheels India	Oracle EBS / SAP	In-house layer	Robotic-weld PLC loops	Custom validation	Manual travel-card scanning; WIP lag
Turbo Energy	Unified ERP	Integrated MES (assembly traceability)	Siemens adaptive CNC controllers	Teamcenter	Controller data isolated from casting-quality engines
Axles India	QAD / SAP	Minimal (siloe logs)	Localised weld PLC	SolidWorks / CATIA	Paper scrap/downtime logs uploaded after shifts
ABI-Showatech	Standalone ERP	Localised batch MES	Casting SCADA	Specialised CAD	NDT records siloe from furnace data
Sundaram Composites	PKC-customised ERP	Manual logging	None (standalone)	Mould-analysis tools	Autoclave cure params trapped in machine software
BIADS	ERP implementation phase	Greenfield MES	PLC-monitored lines	Shared with BIPL R&D	Initial install; high reliance on BIPL IT
Sundaram Dynacast	Proprietary ERP	Manual charts	Multi-slide control cards	Tool-design tools	Shot-speed/die-wear logged on paper

C1.5 Existing AI / automation footprint (do-not-duplicate)

- **Automated foundry (BIPL & ABI):** DISA-partnered lines with computerised sand prep, automatic moulding-density control, laser-guided pouring; high-frequency logging to local historians; plant availability >90%.
- **Adaptive CNC feed-rate control (TEPL):** Siemens **ACM** edge-ML controller on multiple lines — **6-12% cycle-time reduction, 487 machining hours saved in six months, ~20% tool-cost cut** via automated tool-wear monitoring. This is the proven internal blueprint for cluster-wide machining AI.
- **Robotic welding & seam tracking (WIL & AIL):** MIG/TIG cells with automatic weld-gap detection and mechanical seam-tracking.
- **Greenfield mechatronic assembly (BIADS):** automated pick-and-place robotics, computerised torque drivers, automated leak testing for ESC/EPB.

C1.6 AI-readiness ranking (WS1 scorecard, 1-10)

Rank	Entity	Data	Infra	Process	Urgency	Weighted	Strategic AI target
1	Turbo Energy	9	9	9	8	8.75	Predictive assembly-vibration engines from edge controllers
2	Brakes India	8	9	9	8	8.50	CV surface inspection + casting-yield optimisation
3	Wheels India	7	7	8	7	7.25	Predictive maintenance on stamping & robotic-weld cells
4	ABI-Showatech	7	7	9	6	7.25	Casting histories + automated NDT
5	Axles India	6	6	7	6	6.25	Edge-AI weld quality & defect tracking
6	Sundaram Dynacast	5	6	8	5	6.00	ML predictive tool-wear on zinc multi-slide
7	Leggero	5	5	6	5	5.25	Basic digitisation; autoclave cure ↔ part serials
8	BIADS	4	5	6	5	5.00	Native greenfield data schemas

C1.7 Certifications, quality and ESG context

Quality/sector certifications: **IATF 16949:2016** (BIPL, TEPL, WIL, AIL, ABI), **AS9100 Rev D** (ABI Pulivalam aerospace), **TISAX** (BIPL, for European-OEM secure data sharing), **ISO 14001 & 45001** (BIPL, AIL, ABI), **ISO 50001:2018** energy management (BIPL, AIL, ABI). Safety-critical braking/axle/wheel assemblies must comply with **BIS** directives and **AIS** validated by **ARAI** — so any AI used for validation/testing must maintain audit-grade data lineage. ESG: Brakes India’s **Naidupet foundry achieved the world’s first “Green Casting” (CII GreenPro) certification**; renewables ~33% of consumption (FY24); circular use of STP/foundry waste; RO + electrocoagulation cut sludge ~40%.

C1.8 Function-by-function implications (leadership)

Brakes India — *MD (Sriram Viji)*: balance braking dominance with EV/hybrid mechatronics; compress NPD 18→<10 months. *CFO (Shankar)*: margins 13% FY25 (14.7% FY24) — AI demand-forecast + safety-stock optimisation to release working capital; quality analytics to cut cost-of-quality/warranty. *CIDO (Ramya Sampathkumar)*: unify industrial data schema across S/4HANA migrations; shared group AI platform to prevent duplicative builds. *Group President (Vasudevan)*: digital-twin/virtual simulation for ADB/e-park validation under AS9100/ARAI. *President-Foundry (Balaji Krishna)*: ML correlation of mould moisture/pouring speed/chemistry for yield, melt-loss and energy. *LVBU/HVBU heads (Krishnan P / Bheemsingh M)*: predictive assembly diagnostics (torque-driver drift, vibration profiling). *SVP-Aftermarket (Sujit Nayak)*: ML demand forecasting across the distributor network.

Wheels India — *MD (Srivats Ram)*: capex-efficiency lever; *EMCE President (Suresh S)*: ML dimensional-drift analysis on high-tonnage presses; *CFO (Ramesh)*: working-capital/AR analytics against 3.4x leverage; *Ops (Rajeev Kapoor)*: energy minimisation in paint lines, predictive maintenance on weld cells.

Turbo Energy — *Exec Chairman (Anant Ramanujam)*: AI-driven design-space exploration for hydrogen compressors; *ED & CFO (Hayagreevan)*: scale the proven Siemens ACM model across all CNC divisions; *EDs (Operations/Engineering)*: predictive spindle-vibration, automated turbine-wheel vision inspection, generative sand-core prototyping.

Axles India — *MD (Madhavan Varadarajan)*: unified cloud SCADA across three hubs; *CFO (Krishnamoorthy Y)*: scrap-reduction + energy-allocation models; *DGM-Product Dev (K. Vijay Anand)*: predictive weld-quality on submerged-arc lines.

C1.9 Cross-entity scaling opportunities

- **Closed-loop casting-to-machining quality (BIPL → ABI → TEPL)**: link foundry mould/pour parameters to downstream CNC results to predict and pre-empt machining-defect leakage, adjusting foundry variables in real time.
- **Group-wide edge-AI machining engine**: standardise the TEPL Siemens-ACM template across BIPL machining, AIL spindle-finishing and ABI aerospace milling for cluster-wide tooling savings.
- **Multi-plant demand & raw-material optimisation**: one AI commodity-forecasting/inventory engine across AIL, WIL and BIPL (steel/scrap/pig-iron indices + consolidated OEM pull-rates) to lower working capital cluster-wide.

C1.10 Cluster milestones

1960 Wheels India · 1962 Brakes India (Lucas JV) · 1981 BIPL foundry / Axles India · 1982 Turbo Energy · 1991 ABI-Showatech · 1993 Sundaram Dynacast · 2001 BIPL foundry Deming Award · 2020 (Nov) MIND S.r.l. partnership · 2022 (Dec) Leggero Chennai plant · 2024 (Jun) BIADS JV · 2024 (Sep) CIDO appointment · 2025 (May) Dana buy-out → 62.98% Axles · 2025 Jamshedpur greenfield · 2026 (Mar) BIADS Shoolagiri launch.

PART C2 — Auto Distribution & Aftermarket

C2.1 Data reality (read first)

Public information on the three in-scope entities is **thin and asymmetric**: IMPAL is data-rich (listed); Sundaram Motors is partly documented (Mercedes-brand sources); Madras Auto Service is barely documented. The central trap is the **“Sundaram Motors” brand split** post the 2022 settlement — the **Mercedes-Benz** dealership (TN/Karnataka) sits with TSSFPL (in scope); the **Sundaram Honda** network sits with the Rajam branch’s TVS Mobility (out). Pin entity identity before attributing any fact.

C2.2 Sector context — the depth engine

- **Industry size**: Indian auto-component industry **₹6.73 lakh crore (US\$80.2bn) in FY25 (+9.6%)**, nearly doubled in five years. Channels: domestic OEM (>50%), exports (~30%), **aftermarket (replacement) ₹99,948 crore FY25 (+6%)**.
- **Aftermarket outlook**: ICRA projects 8-10% CAGR to **~₹1,55,000 crore by FY30** (replacement-demand growth 5-7% FY25, 7-9% FY26). Demand concentrates in drive-transmission/steering, electricals, suspension and braking — exactly IMPAL’s and MAS’s lines.
- **Counterfeit pathology: 30-40% of aftermarket components leak into grey/counterfeit channels** (FICCI ~30%; ACMA 30-35%), revenue leakage >₹30,000 crore/yr; ACMA/FICCI link ~20% of road accidents to counterfeit/substandard parts. Counterfeits cluster in clutch parts, filters, lamps, wipers, bearings, steering arms and even safety-critical brakes/linings — making the genuine-parts trust premium IMPAL’s and MAS’s moat (peers deploy non-clonable QR and blockchain-style traceability).
- **Workshops**: ~3 lakh fragmented independent workshops vs a small authorised network (~7,000-10,000) — the entire thesis of the aggregator movement.
- **Dealer economics: fixed operations (service + parts) ≈ 49.6% of dealer gross profit** (NADA benchmark) vs new-car net 1-3%; service-and-parts margins 45-55%. For Sundaram Motors-Mercedes, durable value sits in workshop throughput, parts attach and lifecycle retention.
- **Used-car shift**: US\$40-45bn market; **5,412,945 used units vs 4,163,804 new in 2024** (used > new), heading toward 1.7:1 by 2030; organised retail 30%→50% by 2030; **GST on used-vehicle dealer margins raised 12%→18% (Dec 2024)** — pressuring CPO economics.
- **Digital aggregation (cautionary): GoMechanic** (US\$285m valuation, ~1,000-1,200 workshops) admitted fabricated financials Jan 2023, laid off ~70% of staff, sold to Servizzy (Mar 2023); **Spinny acquired GoMechanic Dec 2025** with a US\$160m round. **Boodmo** built an online genuine/aftermarket-parts marketplace; mixed reviews underscore that fulfilment, not catalogue size, is the hard part.
- **Regulatory/EV**: GST (service/parts + used-car hike); vehicle-scrappage policy (uptake may lag); **EV transition** is the largest long-run risk — fewer parts, less maintenance, “engine and transmission product sales could be impacted” (ICRA). Offsets: premiumisation, electronics/safety content, telematics, predictive maintenance, EV-specific components.

C2.3 Per-entity footprint profiles

IMPAL (anchor; listed). Identity: multi-brand auto-parts distributor; BSE 590065 / NSE IMPAL; incorporated 12 Jul 1954; HQ Sundaram Towers, 46 Whites Road, Royapettah, Chennai. Ownership: TSSFPL / TSF Group; SFHL held ~20% (Dec 2023). Represents **40-plus (site “over 50”) component manufacturers** incl. Brakes India, Rane (Madras), Sundram Fasteners, ZF CV Control Systems. Network: **~89 branches / 20 states / 3 UTs** (post CAPL Motor Parts amalgamation), ~18,000 dealers, hub-branch-dealer structure, asset-light pure-trading (capped pricing power). **FY25 sales ₹788.81 crore (+6.43%); net profit ₹83.67 crore (+13.19%); operating margin ~7.5%; net-debt-negative; ~60% of sales from CV spares**. Leadership: **MD Mukund S. Raghavan (DIN 03411396), eff. 5 Jul 2025** (prev. Dy MD); N. Krishnan → Whole-Time Director same date; Chairman S. Ram (non-executive). Digital: B2B dealer-ordering platform launched FY24-25 (inferred); diversifying into EV components (cooling systems, sensors).

Sundaram Motors (TSSFPL / Mercedes-Benz — IN SCOPE). Authorised Mercedes-Benz passenger-car dealer in TN and Karnataka; ~9-10 facilities across ~6 cities (Bengaluru, Chennai, Mangalore, Coimbatore, Salem, Madurai) — the densest Mercedes network in the south; latest facility near Anna Nagar, Chennai. Founded 13 Aug 1945 as a TVS & Sons dealership; Mercedes franchise since 2001 (Karnataka) / 2002-03 (TN). Leadership: ED Sharath Vijayraghavan (DIN 00243575); Brand CEO – Sundaram Group S.S. Ramasubramaniam. Runs Mercedes-mandated systems (Mercedes-Benz India + Zoho **“SKYLine”** DMS; brand-level online booking). Employee counts in third-party databases (~127-347) are low-reliability.

Madras Auto Service (MAS). ~75-branch multi-brand parts wholesaler (south/west/north India); legacy “7,500 dealers” figure is dated. In the TVS fold since 1936; historically India’s largest GM distributor in the 1950s. No named EVP/CEO in public sources (gap). No public DMS/e-commerce/AI evidence. Note duality: a legacy parts/wholesale business also runs on the Rajam side as “TVS Mobility OES & Lubes” — the in-scope MAS is the TSSFPL division (mastvs.com, ~75 branches); branch-level allocation between branches is ambiguous (verify). A “Madras Auto Service” on tradeindia naming “Narayanan Ramji” is a different, unrelated two-wheeler dealer — do not conflate.

C2.4 Customer & operational pain map

Pain	Applies to	Type	Confidence
Right-part-right-branch across a wide multi-state network	IMPAL, MAS	Customer + Ops	Inferred
Counterfeit/grey-market undercutting the genuine-parts proposition	IMPAL, MAS	Customer + Ops	Inferred (sector Verified)
Demand forecasting & inventory across thousands of SKUs / 18,000 dealers	IMPAL	Ops	Inferred
Working-capital / receivables in credit-extended distribution	IMPAL, MAS	Ops	Inferred
CV-cycle exposure (~60% of IMPAL sales) → demand volatility	IMPAL	Ops	Verified
Service-bay throughput, technician capacity	Sundaram Motors	Customer + Ops	Inferred
Showroom/digital booking, lead-to-sale conversion	Sundaram Motors	Customer	Inferred
Used-car (CPO)	Sundaram Motors	Ops	Inferred

sourcing/valuation/margin under 18% GST			
Pricing transparency / repair trust	Sundaram Motors, MAS	Customer	Inferred
EV transition eroding engine/transmission parts (long-term)	IMPAL, MAS	Strategic	Inferred
Digital booking/ordering gaps vs digital-native peers	All three	Ops	Inferred

C2.5 Digital maturity & benchmarks

Entity-level public visibility is minimal: IMPAL’s only documented initiative is the FY24-25 B2B ordering platform (1990s inventory computerisation aside); Sundaram Motors runs OEM-mandated SKYLine DMS; MAS shows no public DMS/e-commerce/CRM. The sector baseline is shifting to cloud DMS, GST-compliant invoicing, omnichannel booking, regional-language interfaces, mobile dealer apps, QR-based parts authentication and AI-driven demand/dealer-performance forecasting — the maturity gap is the central immersion diagnostic.

Family-adjacent benchmark (not in scope): the Rajam-branch **myTVS / Ki Mobility Solutions** is a sophisticated AI-led aftermarket platform — cloud-AI predictive service, “Drivenostics” AI diagnostics, “Life360” connected-car super-app, garage quick-commerce dark stores, a ~1.2-crore-SKU digital parts catalogue, ~10 million customers, guided to ~₹2,100 crore revenue in FY26 and EBITDA-positive. The 2026 TSSF-side JVs with Toyota Kirloskar (**“TSERV Select”** multi-brand service launched Bengaluru 29 Jan 2026; **“UTrust”** pre-owned 17 Apr 2026) show the Santhanam branch itself moving into organised multi-brand service and used cars. Use myTVS as “what good looks like,” explicitly as a comparator, not a target imposed on in-scope entities.

PART C3 — Financial Services: Lending

Scope: Sundaram Finance Ltd (listed deposit-taking NBFC) and Sundaram Home Finance Ltd (HFC subsidiary). Excludes captive two-wheeler financing, insurance, AMC/wealth.

C3.1 Franchise and asset mix

The lending franchise is **branch-led, relationship-heavy and digitally augmenting rather than replacing the field** — “high-touch and high-tech.” Sundaram Finance is overwhelmingly **asset-backed vehicle and equipment finance**. FY25 standalone asset-class concentration: **commercial vehicles 46.12%, cars 26.14%, construction equipment 11.03%, tractors/farm 7.27%, others 9.44%** (“others” ≈ SME, leasing, fuel/tyre working-capital, NBFC finance — not separately disaggregated). On the consolidated view, **housing finance 15.31%** and **non-housing mortgage 8.22%** — meaningful to the group but not the core of the listed parent’s standalone book. The near-term AI value pool therefore sits in relationship-led, collateral-backed, field-heavy underwriting and collections, not pure unsecured-digital lending.

C3.2 Products, ticket and tenor

Sundaram Finance markets car finance, CV finance, construction-equipment finance, farm-equipment finance, diesel/fuel finance, tyre finance, SME finance, leasing and NBFC finance. The financed asset is the primary collateral. New CV finance covers 1-tonne carriers to 49-tonne trailers, up to 60 months; new CE up to 60 months (used CE up to 36); car finance 12–60 months, up to 100% ex-showroom for new cars; farm-equipment finance covers new/used tractors, harvesters and implements with monthly-to-half-yearly repayment; fuel/diesel finance is a card-based working-capital facility (up to 45 days’ credit). Median ticket sizes by product are not publicly disclosed.

Sundaram Home Finance products: home loans, home extension/improvement, plot loans, LAP, commercial-property loans, CRE; the emerging-business unit offers **Anugraha** affordable home loans, **Anugraha-LAP** and **Nidhi** LAP. APR cards distinguish Housing-Salaried, Housing-Self-employed-Individual and Non-Housing. EMI calculator spans ₹1 lakh–₹4 crore and 1–25 years (LAP/home-improvement up to 15 years, home-extension up to 20).

C3.3 Network, hierarchy and channels

Sundaram Finance: **600 branches** nationally. Sundaram Home: **115 branches**, with its emerging-business unit alone at **72 branches** across TN, Telangana, AP and Karnataka, targeting self-employed/informal-income borrowers. SF’s FY25 AR names regional/zonal heads for Tamil Nadu, Northern Region, Western Region, MP/Chhattisgarh & Eastern Region, Karnataka, Andhra, Kerala, Rajasthan, Chennai Zone, Salem & Dharmapuri Zone, plus functional heads for Distribution, Leasing & Corporate Accounts Group, SME & SCF, Collections & Recoveries and HO Operations. Origination is branch-led plus channel partners (SHF’s online business-origination system supports sales/RM/customer/channel-partner workflows; a Live DSA/DMA list is published). Channel-mix percentages are not public.

C3.4 Funding, asset quality and controls

FY26 audited standalone: **deposits ₹8,873.04 crore**, debt securities ₹26,859.79 crore, other borrowings ₹31,261.23 crore; digital deposit opening via online portals and the **SF Next** app. SHF also takes deposits (FY25 deposit base back-solved to ~₹2,345 crore — Inferred). Asset quality is strong: SF **net stage 3 loans 0.69% (FY26)** vs stage-3 assets 1.44% (FY25 highlights); SHF **gross stage III 1.11% / net 0.51% (FY26)**. SHF runs a **Risk Containment Unit** (underwriting-stage screening/sampling) and an **Early Warning Signal** framework (ongoing monitoring). SHF was penalised by RBI in 2024 for non-compliance in periodic KYC updation for certain customers.

C3.5 Existing AI/digital (do-not-duplicate)

Sundaram Finance has already **built** a strong digital foundation and is **scaling** intelligence: a **cloud-based data lake** as the base of a “Systems of Intelligence” layer, **more ML models, agentic-framework pilots**, straight-through processing, digitally guided workflows, task management, a communication framework, embedded nano-learning, customer portals, digital deposits, and a 24x7 SOC. SHF shows more **process digitisation** than explicit AI: online business-origination system, document upload, e-NACH, quick pay, deposit/loan logins, RCU and EWS. Public evidence of chatbots, named RPA, or production propensity models at the lenders was not found. **Workforce:** SF **4,791 permanent employees** (FY25); SHF **1,838** (FY25-end, expansion driven by emerging-business growth). **SHF growth (FY26):** AUM **₹19,909 crore**, disbursements **₹6,842 crore**, EB disbursements **₹590 crore**, with management pointing to Tier-4/Tier-5 town growth.

C3.6 Regulatory frame

Under RBI’s scale-based regulation, **SF = NBFC-Middle Layer** and **SHF = HFC-Middle Layer** (neither on RBI’s January 2025 Upper Layer list). Binding frameworks: **RBI Digital Lending Directions 2025** (RE-LSP arrangements, India-only data storage), updated **KYC Direction**, the 2022 recovery-agent/outsourcing circular, **RBI IT Governance Directions 2023**, IT/NBFC outsourcing directions, and **RBI HFC Directions 2021** for SHF (NHB still visible in complaint escalation). These map directly onto channel, KYC, underwriting, disbursal, servicing and collections stages.

C3.7 Function-by-function implications

Role cluster	Operating tension	Highest-confidence AI implication
EVC / MD / Dy MD / CFO	Growth vs control; funding mix; deposit-franchise durability	AI as an operating-discipline and productivity lever, not front-end novelty
CTDO / digital leadership	Existing data lake, ML, agentic pilots, cyber obligations	Orchestration, monitoring and safe integration into in-motion branch/field workflows
Chief Credit Officer / underwriting	Judgment-heavy, collateral-led appraisal; heterogeneous files; informal-income expansion	Assisted appraisal, document/income interpretation, exception summarisation, controlled override visibility
Regional & zonal heads	Network variation, channel heterogeneity, service consistency, collections productivity	Local prioritisation and workflow support — not stripping decision rights from the field
Head – Collections & Recoveries	Preserve asset quality while scaling/digitising	Early-warning triage, field-route prioritisation, promise-to-pay monitoring, legal-pack readiness

Head – SME & SCF; Leasing & CAG	Cash-flow heterogeneity, document diversity, partner complexity	Document normalisation, covenant/invoice summarisation, RM productivity
SHF MD / CBO / COO	Scale emerging business without diluting quality	Standardise informal-income underwriting, shorten TAT without losing explainability

The AI agenda is augmentation, not autonomous approval — preserving the trusted, field-informed judgment that is a core source of franchise strength.

PART C4 — Financial Services: Investments & Insurance

Scope: Sundaram Asset Management Company, Sundaram Alternate Assets, Royal Sundaram General Insurance.

C4.1 Sector context

- **Mutual funds:** Indian industry **₹65.74 lakh crore AUM (Mar 2025)** (from ₹53.40 lakh crore Mar 2024), 23.45 crore folios; individuals held 60.4% of industry assets, institutions 39.6%.
- **Alternatives:** Indian AIF industry — **₹15.74 lakh crore commitments**, ₹6.79 lakh crore raised, ₹6.45 lakh crore invested; Category II largest by commitments, with a substantial Category III presence.
- **PMS:** SEBI PMS statistics are dominated by EPFO/PF mandates (₹26,98,221 crore vs ₹7,17,171 crore non-EPFO in Jan 2025) — aggregate PMS numbers are not a clean benchmark for a private-client PMS house.
- **General insurance:** non-life industry GDPI **₹3,07,659 crore (FY25)** vs ₹2,89,169 crore (FY24); private-player share eased 53.5%→52.9%; health and motor are the principal growth engines; customer contactability and CKYC refresh are operational priorities.

C4.2 Sundaram AMC

Wholly-owned investment manager to Sundaram Mutual Fund. **Public site: 3.5 million+ investors, 80+ branches, ₹85,000 crore+ AUM.** FY25 AR: **consolidated average AUM ₹76,008 crore**, average mutual-fund AUM **₹64,746.18 crore**, net AUM **₹63,217.14 crore** (31 Mar 2025). Scheme architecture spans equity, thematic, passive/index, hybrid, debt, liquid and FoF (a low-30s live/open-ended set). Distribution: **71,816 empanelled distributors, 79 branches, 712+ Sundaram Finance Group locations**, with **KFin-backed onboarding/KYC** and **MFU/BSE Star MF/NSE NMF** rails; the **InvWise/Enhanced InWise** distributor platform advertises real-time AUM insights, NFO/SIP dashboards, enhanced reports, digital mandates and non-financial transactions. Retail is core; institutional and affluent/intermediated segments are material but not separately broken out. **No clear public entity-level AI deployment claim** (treat “AI in production at AMC” as unverified). Leadership: MD **Anand Radhakrishnan** (CEO Feb 2024 → MD Jun 2024); CBO **C.M. Loganathan**; CFO **R.S. Raghunathan**; Chief Business Operations Officer **H. Lakshmi**; senior fund managers incl. **S. Bharath**, Dwijendra Srivastava.

C4.3 Sundaram Alternate Assets

Wholly-owned subsidiary of Sundaram AMC; PMS + AIF manager. **PMS registration INP000006271**; AIF registrations for the Sundaram Alternative Investment Trust (Cat III) and Sundaram Category II Alternative Investment Trust (Cat II). FY25 AR: **4 Category III + 5 Category II AIF funds, aggregate AIF AUM ₹3,284 crore** (31 Mar 2025). Website: **~₹7,000 crore+ AUM, ~3,000 active customers**, target clientele HNIs/UHNIs/family offices/institutions. Latest granular public PMS data (Mar 2023): 2,744 discretionary clients / ₹2,101.58 crore + 2 advisory clients / ₹249.11 crore. Advisor-led/business-partner/direct model with **digital onboarding, client/distributor logins, PMS-communication digitisation, CKYC, UPI**. No explicit public AI deployment claim. Leadership: MD **Karthik Athreya** (private-credit focus); fund managers incl. Arjun Sankar, Manoj Mahadevan, Darshan Engineer; Zonal Heads **Vinay Vinod Singhi** (South & East) and Ashish Gulati (North).

C4.4 Royal Sundaram General Insurance

Broad general insurer — **motor, health, personal accident, home, travel** (individual) and **fire, marine, engineering, liability, business interruption** (commercial), plus SME and rural products. FY25: **GDP ₹3,763 crore, 27.39 lakh policies issued, 3.68 lakh claims registered**; personal-insurance GWP ₹1,447 crore, commercial GWP ₹1,182 crore, commercial-motor segment premium ₹1,414 crore (labels overlap; a clean retail/commercial split is not publicly reconstructable). Dual operating model with **separate digital environments: VConnect** (retail self-service), **iLink Agent Portal / iLounge** (agent-facing), **commercial portals** (commercial workflows), plus eInsure, InsureX, IRCTC via Insillion, and **FirstGen v7 + NIP v3 on private cloud (FirstApex, Sep 2025)**. Most AI-explicit of the three: early AI pilots, **ML + RPA** in operations and claims, omni-channel claims (bots/WhatsApp/SMS/email/app/web), 2026 brand refresh. Leadership: MD **Vedananarayanan Seshadri** (eff. 28 May 2025); COO **T.M. Shyamsunder**; CFO **Vaibhav Kabra**; CDO-Retail **Ramesh Babu** (trade-reported, lower-confidence); CDO-Commercial publicly unconfirmed (role plausible from portal split).

C4.5 Recurring pain and existing-footprint asymmetry

The recurring friction is **operational complexity under tight regulatory control**, not technology absence: - **AMC:** fund accounting, **valuation/NAV strike**, distributor-channel reconciliation, investor-servicing controls. - **Alternates:** onboarding/KYC/accreditation, closed-end fund administration, capital-call/distribution reporting, PMS/AIF conflict management. - **Insurer:** underwriting discipline, high-volume **claims orchestration**, intermediary/channel management, multi-portal servicing consistency.

Existing-AI evidence is asymmetric: **Royal Sundaram** clearly strongest (ML, RPA, early AI pilots); **Alternates** claims digital onboarding/PMS-comms digitisation; **AMC** has extensive digital servicing/distributor tooling but no explicit production-AI claim. “AI already in place” is far more defensible for the insurer than for the AMC or Alternates.

C4.6 Regulatory frame

SEBI Master Circulars for Mutual Funds (Jun 2024), AIFs (May 2024) and Portfolio Managers (Jul 2025) + Portfolio Managers Regulations 2020 (amended Feb 2025) + **CSCRF (Aug 2024)** for AMCs; IRDAI Insurance Products Regulations 2024, Master Circular on General Insurance Business, Protection of Policyholders’ Interests (Sep 2024), Maintenance of Information Regulations 2025 and **Information & Cyber Security Guidelines 2026** (6-hour incident reporting). Any AI/data layer interfacing with these operations inherits a heavier control burden — especially where it touches NAV, suitability, underwriting, claims or customer records.

PART C5 — Shared Services & Social

Scope: Sundaram Business Services (SBS), Sundaram Medical Foundation (SMF), Laxmi Charities.

C5.1 Sector context

- **BPM/shared services:** Indian sector US\$250bn+, shifting from T&M/cost-plus to transaction-based/outcome-linked pricing; **NASSCOM-EY AI maturity 2.47/4 (“Enthusiast”)**; an **“agentic shift”** away from rule-based RPA toward gen-AI-human collaboration; contact-centre voice revenue projected to fall 50-60%→30-40% over three years. Barriers: legacy/fragmented architectures, scarce standardised compliant datasets, and a shortage of domain+AI+prompt talent.
- **Not-for-profit healthcare:** cross-subsidisation model; governed by **ABDM** (860M+ ABHA IDs, consent-based federated architecture), the **Digital Health Incentive Scheme** (up to ₹5 crore for KYC-verified ABHA-linked records), the 2026 **SAHI** strategy (AI augments, not replaces, clinical judgment), and the **BODH** validation sandbox. Healthcare-IT market US\$19.36bn (2025) → US\$112.09bn (2034) at 19.7% CAGR; cloud 57.6% of deployments. Indian consumer AI readiness high (78% use gen-AI to understand diagnoses); hospitals cautious (>50% on ad-hoc AI budgets, focused on radiology/pathology and ambient scribing).
- **CSR/charitable trusts:** Companies Act s.135 (2% of average net profit; thresholds net worth ≥₹500cr / turnover ≥₹1,000cr / net profit ≥₹5cr, with a proposed move to ₹10cr); **revised Form CSR-1 (eff. 14 Jul 2025)** requiring a 3-year track record, 12A/80G registrations, professional certification; mandatory third-party impact assessments (≥₹1cr projects); strict unspent-CSR rules. These collide with **DPDP** minimisation — the “privacy paradox” of donor reporting vs beneficiary-data protection.

C5.2 Sundaram Business Services (SBS)

Global BPM/outsourcing under SFHL, serving Australia/NZ/UK/India. Services: **Finance & Accounting** (outsourced-CFO, practice support); **SMSF administration** (end-to-end admin + audit support; ~5,000 super funds/yr, 70+ specialists); **PMS platform support / fund administration; mortgage support** (processing, vetting, document validation, compliance for **3,600+ brokers/aggregators/lenders**). 400+ professionals; internal-lineage + external global clients (substantial third-party Australian accounting/wealth book). Mandate shift to **outcome-based / AI-native** delivery and prompt-engineering competency, under strict **client-specific data isolation**.

AI relevance themes: (1) **Agentic AI for SMSF administration & audit reconciliations** — multi-agent orchestrator parsing unstructured PDFs in client cloud platforms (e.g. Xero), matching transactions and flagging anomalies; potential ~80% reduction in manual account-prep effort. (2) **IDP for mortgage underwriting support** — vision-language IDP across payslips/bank statements/tax returns/ID, cross-referencing lending policy; turnaround days → hours.

Digital maturity: NASSCOM “Expert” (3.10). Client-isolated private VLANs + Active Directory; secure RDP/cloud-API linkages (process on client-hosted servers, **zero local data residency**); ISO 27001 + ISO 9001; intranet QMS portal; LMS.

C5.3 Sundaram Medical Foundation (SMF)

Public not-for-profit multi-specialty hospital (founded 1990, Dr S. Rangarajan). **230+ beds**, Anna Nagar + Mogappair; inpatient/outpatient across **30+ clinical specialties**; postgraduate **DNB** teaching (NBE-accredited, 8 specialties); community health/outreach incl. STEP. Cross-subsidy: **30% free outpatients / 15% free inpatients**.

AI relevance themes: (1) **Ambient voice scribing + clinical decision support** — consented ambient capture → medical-LLM transcription → HL7 FHIR notes for clinician sign-off; potential 35-45% cut in documentation time. (2) **Conversational AI / virtual triage for women’s health** — scaling the clinically-validated **“Dear Akka”** companion (with CAN-STOP) from a manual phone helpline (044 3569 3070) to a WhatsApp assistant guiding Tamil/English symptom assessment across seven conditions (incl. breast/cervical/ovarian cancers) with risk-based referral.

Digital maturity: NASSCOM “Enthusiast” (2.50). Centralised HIS + UHID registry; in-house “Digicare”/Digital Health teams; transitioning to **ABDM v3 / FHIR**; centralised dispatch/booking linked to UHID. Constraints: DPDP (health data inherently high-risk), HL7 FHIR interoperability, and **ICMR Ethical Guidelines for AI in Healthcare (2023)** requiring human oversight, transparency and fairness.

C5.4 Laxmi Charities

Public charitable trust (1973), trustees from the Sundaram Finance/Brakes India/Turbo Energy lineages. Remit: **scholarships** (school→higher education), the **Vridhi Functional English** initiative (with Sundaram Learning Centre), and structured **mentorship** pairing recipients with senior group leaders. **200,000+ students supported** over five decades; ~two-thirds female; 50%+ from families earning <₹10,000/month. Disbursements are **institution-routed**, not direct to individuals.

AI relevance themes: (1) **Predictive attrition modelling & mentorship allocation** — ML over the CMI-built tracking database (academic trends, attendance, socio-economic variables) to flag at-risk students early and route mentors. (2) **Generative-AI English tutors for Vridhi** — on-demand conversational practice with Tamil-supported explanations, scaling beyond volunteer-trainer constraints.

Digital maturity: NASSCOM “Explorer” (1.80). Primary asset is the **CMI student-tracking database**; secure online scholarship-application/nomination portal; interactive mentorship forms. Governance: DPDP as Data Fiduciary, **verifiable parental consent** for minors, beneficiary-identifier minimisation, and a “trust-based philanthropy” posture (qualitative/aggregated impact over detailed PII) to reconcile donor reporting with privacy.

C5.5 Cross-entity enablement logic

A unified data/technology framework and a central DPO function could modernise three distinct models at once: SBS’s transactional expertise can streamline SMF billing and Laxmi scholarship administration, while a shared responsible-AI and consent architecture covers all three under DPDP. This is the structural rationale for grouping these CEOs with the group digital mandates in the enablement table (Part D, T9).

PART D — Participant Clustering (nine working tables)

Seventy-eight leaders organised via a generate → filter → classify → verify workflow into **nine tables** — six domain-deep, two cross-functional (T8 Finance, T9 Enablement), one auto-distribution (T4). Anchor = most senior/relevant leader (*). All 78 placed exactly once; Sundaram Finance’s 18 split four ways (T5 credit, T6 collections, T8 finance, T9 digital/board); no orphaned singletons.

T1 — Precision Braking & Foundry Systems (9; domain-deep)

Territory: braking-system design, foundry yield, casting-defect prediction. *Anchor:* Sriram Viji (Brakes India, MD). Members: Sriram Viji*, Shankar (CFO), Madhu (ED Corporate Affairs), Vasudevan (Group President), Balaji Krishna (President-Foundry), Krishnan P (President-LVBU), Bheemsingh M (President-HVBU), Sujit Nayak (SVP-AMBU) — all Brakes India; Tirumalai (COO, Sundaram Composites).

T2 — Mobility Components & Wheels (9; domain-deep)

Territory: wheel/component manufacturing, plant operations, demand-linked sales. *Anchor:* Srivats Ram (Wheels India, Chairman & MD). Members: Srivats Ram*, Sanjay Pande (SVP Marketing), Murali Vaidyanathan (President-Automotive), Suresh S (President-EMCE), Ranjan Ambirajan (Executive-Sales), Rajeev Kapoor (Head-Operations), Mukundanarayanan (SGM-Accounts) — all Wheels India; KVP Krishnadas (Head, BIADS); Yuvaraj (Head, Sundaram Dynacast).

T3 — Powertrain & Driveline Systems (9; domain-deep)

Territory: turbocharger/axle product development, testing, assembly quality. *Anchor:* Anant Ramanujam (Turbo Energy, Executive Chairman). Members: Anant Ramanujam*, M Vanamurthy (VP), R Balaji (VP), Ashok Kumar Yadav (VP), Ravindran (ED) — Turbo Energy; Madhavan Varadarajan (MD), K. Vijay Anand (DGM-Product Dev) — Axles India; Aditya Ramanujam (ED), Srikant Ramanjum (MD) — ABI-Showatech.

T4 — Aftermarket & Vehicle Retail (6; domain-deep)

Territory: parts distribution, dealership retail, service, spares forecasting. *Anchor:* Mukund Raghavan (IMPAL, MD). Members: Mukund Raghavan*; Sharath Vijayaraghavan (ED), SS Ram (EVP-Cars), Suryanarayanan D (SGM-Finance) — Sundaram Motors; Gopal Maloo (EVP), Jacob (CEO) — Madras Auto Service.

T5 — Lending & Credit Decisioning (9; domain-deep)

Territory: underwriting, origination, SME/home-loan decisioning, pricing. *Anchor:* Rajiv C Lochan (Sundaram Finance, MD). Members: Rajiv C Lochan*, Raju AN (Deputy MD), Rahul Jyothi Kumaran (Head SME & SCF), Narayanan K (Head Leasing & CAG), Muralidharan V (Head S&P) — Sundaram Finance; Laks (MD), Nishant P (CBO), Hema Ganapathy (COO), Kumaresh Balasubramanian (GM-Projects) — Sundaram Home Finance.

T6 — Collections & Branch Network (8; domain-deep)

Territory: collections, regional branch productivity, distribution-network optimisation. *Anchor:* Rajesh Venkat (Sundaram Finance, Head TN). Members (all Sundaram Finance): Rajesh Venkat*, Salin Nair (Head North), Sunil S Narayanan (Head MP/Chhattisgarh & East), Manoj Nair (Head Distribution), Tom Thomas (Head West), Jeevan Jose (Zonal Head Salem & Dharmapuri), Sri Raman (Dy Region Head Telangana), Shankar Thiagarajan (Dy Head Karnataka).

T7 — Investments, Wealth & Insurance (11; domain-deep, at size cap)

Territory: fund management, wealth advisory, underwriting, claims. *Co-anchors:* Anand Radhakrishnan (Sundaram AMC, MD) / Vedanarayanan Seshadri (Royal Sundaram, MD). Members: Anand Radhakrishnan*, Loganathan CM (CBO), Bharath S (Sr Fund Manager), Lakshmi H (Chief Business Operations Officer) — Sundaram AMC; Karthik Athreya (MD), Arjun Sankar (Fund Manager), Vinay V Singhvi (Zonal Head S&E) — Sundaram Alternates; Vedanarayanan Seshadri, Shyamsunder TM (COO), Sathish Kumar S (CDO Commercial), Ramesh Babu (CDO Retail) — Royal Sundaram.

T8 — Group Finance & Risk (8; cross-functional)

Territory: group financial planning, consolidation, audit, treasury/risk. *Anchor:* Harsha Viji (Sundaram Finance, Executive Vice Chairman). Members: Harsha Viji*, Ramaswamy M (CFO), Bama Balakrishnan (SVP Finance) — Sundaram Finance; Ramesh (CFO, Wheels India), Haygreevan (ED & CFO, Turbo Energy), Krishnamoorthy Y (CFO, Axles India), Vaibhav Kabra (CFO, Royal Sundaram), Raghunathan RS (CFO, Sundaram AMC).

T9 — Group Enablement: Digital, Data, People & Social (9; cross-functional)

Territory: shared AI/data platform and responsible-AI capability; AI for HR, shared services and the social arms. *Apex sponsor:* Sudheer Warriar (Sundaram Finance, CTDO). Members: Sudheer Warriar*, Ramachandran N (ED) — Sundaram Finance; Ramya Sampathkumar (CIDO, Brakes India); Venkataswami Babu (VP HR, Wheels India); R. Badrinath (GM HR, Axles India); Ganapathi Subramanian S (CHRO, Sundaram Home Finance); Deepa Ramesh (CEO, Sundaram Business Services); Arjun Rangarajan (Director, Sundaram Medical Foundation); Shyamala Ashok (CEO, Laxmi Charities).

PART E — Challenge Card Portfolio (27 cards)

One primary (P1) and two secondary (S1, S2) per table. Each card states a problem only — no solution. Impact = the deck's 1-5 business-impact rating; anchors quantify the problem's scale.

T1 — Precision Braking & Foundry

T1-P1 · Foundry yield (Impact 5). A third or more of every molten-iron charge leaves the foundry as scrap to remelt, and the conditions for a sound, high-yield pour can't be reliably held. *Why now:* operating margin moderated to ~13% (FY25), focusing conversion cost; induction melting re-incurs ~550-650 kWh/tonne. *Anchors:* 55-65% yield; 35-45% returned to remelt; 2-4% melt-loss/heat; ~2,00,000 t/yr capacity. *Owner:* Balaji Krishna (President, Foundry); Shankar (CFO). **T1-S1 · Casting defect escapes (Impact 4).** Safety-critical casting defects are caught only at end-of-line or escape to the customer, and a defect can't be traced back to the conditions that caused it. *Anchors:* 4-6% internal rejection; OEM sub-50-PPM targets; shift-decay in manual inspection. *Owner:* Balaji Krishna; Vasudevan (Group President); Sriram Viji (MD). **T1-S2 · 12-18 month NPD loop (Impact 4).** A new braking component takes 12-18 months to market because each design crawls through a sequential physical loop of tooling, prototypes, machining and endurance testing. *Why now:* MD ambition to compress NPD toward <10 months; expressways and rising axle loads drive advanced-braking demand. *Owner:* Sriram Viji; Vasudevan; Tirumalai (COO, Sundaram Composites).

T2 — Mobility Components & Wheels

T2-P1 · Demand & working capital (Impact 5). Abrupt OEM pull-rate swings outrun static forecasting, whipsawing steel and WIP inventory and trapping working capital. *Owner:* Sanjay Pande; Ranjan Ambirajan; Rajeev Kapoor; Mukundanarayanan. **T2-S1 · CNC tool life (Impact 4).** The fixed-schedule-versus-run-to-failure tooling trade-off on CNC rim machining either bins tools with life left or scraps the workpiece (20-30% useful life lost on fixed swaps). *Owner:* Rajeev Kapoor; Suresh S; Murali Vaidyanathan. **T2-S2 · Stamping dimensional drift (Impact 4).** Dimensional drift on high-tonnage EMCE stamping presses pushes parts out of tolerance, sacrificing first-time-right quality. *Owner:* Suresh S; Rajeev Kapoor.

T3 — Powertrain & Driveline

T3-P1 · Turbo unbalance (Impact 5). Turbocharger cartridges are rejected for dynamic unbalance only at end-of-line testing, where the cause can't be pinned to a single upstream step. *Owner:* M Vanamurthy & R Balaji (VPs); Ravindran (ED). **T3-S1 · Aerospace first-article qualification (Impact 4).** Every new aerospace casting must clear a slow, documentation-heavy first-article qualification that gates production release. *Owner:* Srikant Ramanjum (MD); Aditya Ramanujam (ED), ABI-Showatech. **T3-S2 · Axle seam-weld defects (Impact 4).** Seam-weld defects on heavy axle housings drive expensive manual post-weld rework that is detected late and recorded on paper. *Owner:* K. Vijay Anand; Madhavan Varadarajan, Axles India.

T4 — Aftermarket & Vehicle Retail

T4-P1 · Right part, right branch (Impact 5). The correct part is not reliably in the correct branch against fragmented, hyper-local dealer demand (89 branches × ~18,000 dealers). *Owner:* Mukund Raghavan (IMPAL); Gopal Maloo & Jacob (Madras Auto Service). **T4-S1 · Dealership fixed-ops (Impact 4).** Service-bay throughput and ownership-lifecycle retention leak because service demand and customer return are managed without a forward view (fixed-ops ≈ 49.6% of dealer gross profit). *Owner:* Sharath Vijayaraghavan; SS Ram; Suryanarayanan D, Sundaram Motors. **T4-S2 · Genuine-parts provenance (Impact 4).** The genuine-parts trust premium is eroding because genuineness is asserted by reputation but can't be demonstrated at the point of purchase (30-40% grey market). *Owner:* Jacob (MAS); Mukund Raghavan (IMPAL).

T5 — Lending & Credit Decisioning

T5-P1 · Document interpretation (Impact 5). Interpretation of heterogeneous borrower documents is slow and uneven across the branch network, varying decision quality and turnaround (600+ branches). *Owner:* Rahul Jyothi Kumaran; Narayanan K; Rajiv C Lochan. **T5-S1 · Informal-income capacity (Impact 4).** True repayment capacity for self-employed, informal-income borrowers is hard to evidence consistently at scale (the 72-branch emerging-business segment). *Owner:* Nishant P; Hema Ganapathy; Laks, Sundaram Home Finance. **T5-S2 · Pricing consistency (Impact 3).** Borrowers of similar profile, ticket and credit standing receive inconsistent and hard-to-explain pricing outcomes. *Owner:* Hema Ganapathy; Nishant P; Muralidharan V.

T6 — Collections & Branch Network

T6-P1 · Field-collection targeting (Impact 5). Field-collection effort is spread without a reliable view of which overdue accounts most warrant a visit. *Owner:* Rajesh Venkat; Jeevan Jose; Sri Raman. **T6-S1 · Early warning (Impact 4).** Early signs that a paying customer is about to slip surface too late to act softly (vs the existing EWS framework). *Owner:* Shankar Thiagarajan; Salin Nair; Tom Thomas. **T6-S2 · Branch recovery variance (Impact 3).** Comparable branches recover at materially different rates with no agreed explanation. *Owner:* Manoj Nair; Sunil S Narayanan; Rajesh Venkat.

T7 — Investments, Wealth & Insurance

T7-P1 · Claims orchestration (Impact 5). Claims move slowly and unevenly because intake, assessment and settlement are stitched across many channels and product silos (3.68 lakh claims/yr). *Owner:* Shyamsunder TM; Sathish Kumar S; Ramesh Babu, Royal Sundaram. **T7-S1 · NAV-strike reconciliation (Impact 4).** The daily NAV-strike deadline imposes a heavy, error-sensitive valuation and reconciliation load, where any break is investor-visible. *Owner:* Lakshmi H; Bharath S, Sundaram AMC. **T7-S2 · Affluent onboarding (Impact 4).** Onboarding, suitability and ongoing reporting for affluent investors are slow and control-heavy in a partner-led PMS/AIF model. *Owner:* Vinay V Singhvi; Arjun Sankar, Sundaram Alternates.

T8 — Group Finance & Risk

T8-P1 · Group close (Impact 5). The group close depends on stitching together entities on different calendars, ledgers and regulatory definitions before a consolidated number can be trusted. *Owner:* Ramaswamy M; Bama Balakrishnan; Harsha Viji. **T8-S1 · Intercompany elimination (Impact 4).** Intercompany balances must be matched and eliminated before any consolidated number holds, yet each pair records them differently. *Owner:* Krishnamoorthy Y; Ramesh; Bama Balakrishnan. **T8-S2 · Group cash view (Impact 4).** The group has no single, timely view of where its cash sits,

because balances are spread across ring-fenced regulated entities. *Owner:* Vaibhav Kabra; Raghunathan RS; Haygreevan.

T9 — Group Enablement

T9-P1 · Cross-entity master data (Impact 5). The group can't recognise the same customer, supplier or part across its separately-owned companies, because each defines and holds its own master data. *Owner:* Sudheer Warriar (CTDO); Ramya Sampathkumar (CIDO); Ramachandran N (ED). **T9-S1 · Workforce friction (Impact 4).** Workforce friction runs high because the group's very different manufacturing and financial-services talent pools are managed in disconnected HR functions. *Owner:* Venkataswami Babu; R. Badrinath; Ganapathi Subramanian S. **T9-S2 · Shared-services margin (Impact 4).** High-volume transaction processing under strict client-data isolation keeps shared services manual and caps its margin (SMSF ~5,000 funds/yr; mortgage 3,600+ brokers). *Owner:* Deepa Ramesh (CEO, Sundaram Business Services).

PART F — Build-Readiness Assessment

Independent assessment of all 27 challenges on four axes. **AI fit** = how readily AI addresses it today; **Novelty** = how differentiated the AI application is (H = under-exploited/hard, L = commodity pattern); **Usefulness** = workflow value if solved; **Impact** = business impact to the entity. Most challenges are *proven patterns new to this group* rather than technically novel — the right risk profile for a build-in-a-day immersion; only a few are genuinely novel-and-hard.

Ref	Challenge	AI fit	Novel	Useful	Impact	Verdict
T1-P1	Foundry yield / unstable pour	H	M	H	H	Strong; multivariate process control; gated by sensor/data readiness
T1-S1	Casting defect escapes + traceability	H	L-M	H	H	Build-ready; vision commodity, traceability is the value
T1-S2	12-18mo NPD loop	M	M-H	H	H	Most differentiated, least build-ready in a day — roadmap
T2-P1	Demand & working capital	H	L	H	H	Commodity demand-sensing; value in OEM-signal integration
T2-S1	CNC tool life	H	M	M-H	M	Solid predictive-maintenance slice; narrow
T2-S2	Stamping drift	H	L-M	M-H	M	Proven SPC+ML; narrow scope
T3-P1	Turbo unbalance at EOL	H	M	H	H	Strong root-cause slice; data-dependent
T3-S1	Aerospace first-article qualification	M-H	M	M-H	M	AI assists doc burden; aerospace regulatory limits autonomy
T3-S2	Axle seam-weld defects + paper rework	H	L-M	M-H	M	Vision inspection + digitise records; build-ready
T4-P1	Right part, right branch	H	L	H	H	Commodity but high-value distributed forecasting
T4-S1	Dealership fixed-ops	H	L	M-H	M	Proven scheduling/churn models
T4-S2	Genuine-parts provenance	L-M	M-H	M	M	Weakest AI fit — mainly serialisation/anti-counterfeit
T5-P1	Borrower-document interpretation	H	L	H	H	Canonical GenAI/IDP slice; human-in-loop on decisions
T5-S1	Informal-income capacity	M-H	M	H	H	Valuable for India; RBI fair-lending/explainability caution
T5-S2	Pricing consistency	M-H	M	M-H	M	Policy-laden; explainability is the hard part
T6-P1	Field-collection targeting	H	L	H	H	Proven prioritisation; clear ROI
T6-S1	Early warning	H	L-M	H	M-H	Proven behavioural scoring
T6-S2	Branch recovery variance	M	L-M	M	M	More diagnostic BI than AI build
T7-P1	Claims orchestration	H	L-M	H	H	Insurtech-proven triage/automation; IRDAI-aware
T7-S1	NAV-strike reconciliation	M-H	L-M	H	M-H	Break/anomaly detection; very low error tolerance
T7-S2	Affluent onboarding	M-H	L-M	M-H	M-H	KYC + report automation; SEBI suitability constraints
T8-P1	Group close	M	M	H	H	Core is data-integration; AI adds mapping/anomaly/narrative
T8-S1	Intercompany elimination	M-H	L-M	H	M-H	Good transaction-matching slice
T8-S2	Group cash view	L-M	L	M-H	M	Mainly integration + ring-fencing; AI = cash forecasting on top
T9-P1	Cross-entity master data	H	M-H	H	H	Most differentiated; entity resolution is real AI — but DPDP + separate ownership make it an org problem too
T9-S1	Workforce friction	M	L-M	M	M	Diffuse/soft; weakest build

						candidate
T9-S2	Shared-services margin	H	L-M	H	M-H	Directly margin-accretive; data-isolation is a design constraint

F.1 Where to steer attention

- **Strongest build-ready candidates** (high on all four, low caveats): **T1-S1, T3-P1, T5-P1, T6-P1, T7-P1, T9-S2** — safe primaries to actually prototype in the room.
- **Most differentiated but harder: T9-P1** (cross-entity entity resolution — the one genuinely group-unlocking idea), **T1-P1** (foundry pour control), **T1-S2** (NPD compression). Worth doing, but won't finish as a clean slice in a day.
- **Treat carefully — real problem, but not mostly an AI problem: T4-S2** (provenance → serialisation), **T6-S2** (branch variance → diagnostic analytics), **T8-S2** (cash view → integration), **T9-S1** (HR → diffuse/change-management). Nudge these tables toward the AI-tractable slice inside the problem.

F.2 Two cross-cutting build caveats

1. **Manufacturing data readiness.** T1-T3 cards assume usable sensor/process data exists. Confirm instrumentation before promising a working slice, or prototype on synthetic data. The TEPL Siemens-ACM precedent shows where real edge data already exists.
2. **Regulated-decision guardrails.** Every lending/insurance card (T5, T6-S1, T7) touches a regulated decision. Keep human-in-the-loop and explainability explicit (RBI Digital Lending Directions, IRDAI, SEBI), or the prototype isn't deployable.

F.3 Expectation-setting on novelty

If leadership asks “is this cutting-edge?”, the honest answer is that most cards are *proven elsewhere, new to TVS* — exactly the right risk profile for an immersion. Only T9-P1 and the T1 pair are genuinely novel-and-hard. This aligns with the group’s federated posture: finance has the clearest AI ambition, manufacturing the clearest digital substrate, and the cross-entity data problem (T9-P1) is where the highest group-level upside and the highest organisational difficulty coincide.

PART G — Consolidated Gaps, Naming Reconciliations & Confidence Map

G.1 Naming reconciliations (apply consistently across the pack)

- **Sundaram Finance “CTDO”** = *Chief Technology & Digital Officer* (Sudheer Warriar) — **not** “Chief Transformation & Digital Officer.”
- **A.N. Raju** is publicly *Deputy Managing Director*; the “Joint MD” label used in some materials is **unverified**.
- **Ramya Sampathkumar** = *SVP & Chief Information & Digital Officer, Brakes India* (manufacturing-side mandate) — distinct from the SF CTDO role.
- **“Sundaram Motors”** (in-scope, Mercedes-Benz dealership division of TVS & Sons) ≠ **“Sundaram Honda”** (out-of-scope, TVS Mobility / Rajam).
- **“Sundaram Finance Holdings”** ≠ the defunct interim name “Sundaram Finance Investments Ltd” (renamed 2017).
- Wheels India MD is also a director of Axles India and Leggero; treat cross-directorships as governance links, not separate roles.

G.2 Open questions to close live at the immersion

1. Brakes India CIN, exact incorporation date and registered office; Wheels India full equity CIN; IMPAL ISIN.
2. Sundaram Alternate Assets CIN, incorporation date (PMS reg INP000006271 confirmed; AIF trust names confirmed).
3. Exact RBI NBFC CoR (SF) and NHB/RBI CoR (SHF) numbers; whether SFHL is an RBI-registered Core Investment Company.
4. Private-company ownership percentages (Brakes India, Turbo Energy, Sundaram Dynacast, ABI-Showatech, Leggero, BIADS).
5. **Flometallic India Pvt Ltd** — confirm whether it is an in-lineage entity to add to the roster.
6. Named org charts/leadership for Sundaram Motors (EVP-Cars, SGM-Finance) and Madras Auto Service (EVP, CEO) — the largest people-data gaps; Royal Sundaram CDO-Commercial publicly unconfirmed.
7. Real per-entity **digital baselines** (DMS, MES, historians, CRM, e-commerce, analytics) before any solution framing — to avoid duplicating live initiatives (especially SF’s data lake/agentive pilots, Royal Sundaram’s FirstApex, TEPL’s Siemens ACM).
8. SBS internal-vs-external revenue split; SMF IT/clinical-systems estate and daily clinical-transaction volume; Laxmi Charities disbursement scale and audit status.
9. Incorporation-month discrepancies (TVS & Sons Oct vs Dec 1929; Sundaram Finance Aug vs Dec 1954) — reconcile against MCA master.

G.3 Confidence map at a glance

- **Highest (Verified — primary/regulator/exchange):** listed-entity facts (Sundaram Finance, SFHL, Wheels India, IMPAL), regulator mappings, lending asset-class mix and stage ratios, insurer premium/claims volumes, AMC/AIF AUM, the two named digital mandates, group-level regulatory constraints, manufacturing entity financials and the foundry/machining pain metrics, the Siemens-ACM result.
- **Moderate (Inferred):** private-company ownership splits, manufacturing pain magnitudes mapped from sector norms, distribution digital baselines, retail-vs-commercial insurance split, SHF deposit scale, “build tractability highest in manufacturing,” the CIDO remit interpretation.
- **Lowest (Assumed / to confirm):** unincorporated-division and trust particulars, Sundaram Motors/MAS leadership, several dated footprint figures (MAS “7,500 dealers”), AMC production-AI claims, exact private ownership percentages.

G.4 Source families relied upon

MCA/ROC master data (via Zaubacorp/Tofler/Tracxn mirroring MCA21), SEBI/NSE/BSE disclosures (shareholding patterns, Reg.30/31 intimations), NHB HFC list, SEBI Mutual Fund/PMS/AIF registers and master circulars, IRDAI registered-insurer list and circulars, RBI scale-based-regulation framework and Digital Lending / IT Governance directions, audited annual reports (sundaramfinance.in, sundaramholdings.in, Wheels India, IMPAL, Royal Sundaram, Sundaram AMC, Sundaram Home Finance), entity websites and product pages, ACMA/ICRA/FICCI sector data, NASSCOM-EY AI Adoption Index, Bain healthcare reporting, MeitY DPDP materials and MCA CSR notifications, credit-rating rationales (CRISIL/ICRA/India Ratings/CARE), and reputable trade press (Autocar Professional, MotorIndia, Business Line, Industrial Economist, M&A Critique, MarkLines, Times of India) used for corroboration and narrative only — never as the sole basis for an entity-level fact.

End of consolidated knowledge pack (expanded edition). Every workstream’s substantive context is represented here once; lineage rules (Part A) and group AI posture/constraints (Part B) are stated centrally and assumed across the domain deep-dives, the clustering, the challenge portfolio and the build-readiness assessment.

PART H — Consolidated Leadership Directory

Named executives evidenced across the workstreams, by entity. Use with the naming reconciliations in G.1. “(room)” marks leaders attending the immersion (per the clustering).

H.1 Financial services

Entity	Role	Leader
Sundaram Finance	Executive Vice Chairman	Harsha Viji (room)
Sundaram Finance	Managing Director	Rajiv C Lochan (room)
Sundaram Finance	Deputy Managing Director	A.N. Raju (room)
Sundaram Finance	Chief Financial Officer	Ramaswamy M (room)
Sundaram Finance	Executive Director	Ramachandran N (room)
Sundaram Finance	Chief Technology & Digital Officer	Sudheer Warriier (room)
Sundaram Finance	SVP, Finance	Bama Balakrishnan (room)
Sundaram Finance	National Head, Collections & Recoveries	(named in AR; confirm)
Sundaram Finance	Regional/zonal & functional heads	Rajesh Venkat (TN), Salin Nair (North), Sunil S Narayanan (MP/Chh & East), Tom Thomas (West), Manoj Nair (Distribution), Rahul Jyothi Kumaran (SME & SCF), Narayanan K (Leasing & CAG), Muralidharan V (S&P), Jeevan Jose (Salem & Dharmapuri), Sri Raman (Telangana), Shankar Thiagarajan (Karnataka) — all room
Sundaram Home Finance	Managing Director	Laks (room)
Sundaram Home Finance	Chief Business Officer	Nishant P (room)
Sundaram Home Finance	Chief Operating Officer	Hema Ganapathy (room)
Sundaram Home Finance	CHRO	Ganapathi Subramanian S (room)
Sundaram Home Finance	GM, Projects (HD & SD Channels)	Kumaresh Balasubramanian (room)
Sundaram AMC	Managing Director	Anand Radhakrishnan (CEO Feb 2024 → MD Jun 2024) (room)
Sundaram AMC	Chief Business Officer	C.M. Loganathan (room)
Sundaram AMC	Chief Financial Officer	R.S. Raghunathan (room)
Sundaram AMC	Chief Business Operations Officer	H. Lakshmi (room)
Sundaram AMC	Senior Fund Manager	S. Bharath (room); also Dwijendra Srivastava
Sundaram Alternates	Managing Director	Karthik Athreya (room)
Sundaram Alternates	Fund Manager	Arjun Sankar (room); also Manoj Mahadevan, Darshan Engineer
Sundaram Alternates	Zonal Head, South & East	Vinay V Singhvi (room); North: Ashish Gulati
Royal Sundaram	Managing Director	Vedanarayanan Seshadri (eff. 28 May 2025) (room)
Royal Sundaram	Chief Operating Officer	T.M. Shyamsunder (room)
Royal Sundaram	Chief Financial Officer	Vaibhav Kabra (room)
Royal Sundaram	CDO – Retail	Ramesh Babu (trade-reported) (room)
Royal Sundaram	CDO – Commercial	Sathish Kumar S (room; public naming unconfirmed)

H.2 Manufacturing & auto components

Entity	Role	Leader
Brakes India	Managing Director	Sriram Viji (room)
Brakes India	ED & CFO	G. Shankar (room)
Brakes India	Director, Corporate Affairs & HR	R. Madhusudhan (“Madhu”) (room)
Brakes India	SVP & CIDO	Ramya Sampathkumar (room)
Brakes India	Group President (Brakes Business)	K. Vasudevan (room)
Brakes India	President, Metal Castings/Foundry	V. Balajikrishna (room)
Brakes India	President, Light Vehicles BU	P. Krishnan (room)
Brakes India	President, Heavy Vehicles BU	D. Bheemsingh Melchisedec (room)
Brakes India	SVP, Aftermarket BU	S. Sujit Nayak (room)
Wheels India	Chairman & Managing Director	Srivats Ram (room)
Wheels India	CFO	P. Ramesh Srinivas (room)
Wheels India	President, EMCE / Quality & NPD	K. Gnanasubramanian (“Suresh S”) (room)
Wheels India	President-Automotive, SVP-Marketing, Sales, Operations, HR, Accounts	Murali Vaidyanathan, Sanjay Pande, Ranjan Ambirajan, Rajeev Kapoor, Venkataswami Babu, Mukundanarayanan (room)

Turbo Energy	Executive Chairman	Anant Ramanujam / Santhanam Ram (room)
Turbo Energy	ED & CFO	T.T. Hayagreevan (room)
Turbo Energy	ED, Operations	P.S. Dasarathy
Turbo Energy	ED, Engineering & R&D	Dr. D.A. Subramani
Axles India	Chairman	Santhanam Ram
Axles India	Managing Director	Madhavan Varadarajan (room)
Axles India	CFO	Krishnamoorthy Y (room)
Axles India	DGM, Product Development	K. Vijay Anand / Thennavan Venugopal (room)
Axles India	GM, HR	R. Badrinath (room)
ABI-Showatech	Managing Director	Srikant Ramanjum / Ramanujam Srikanth (room)
ABI-Showatech	Executive Director, Operations	Aditya Ramanujam / Sundar K (room)
Sundaram Composites (Leggero)	CEO	Rajiv Balagangadharan
Sundaram Composites (Leggero)	COO	G. Thirumalai (room)
BIADS	Head	KVP Krishnadas (room)
Sundaram Dynacast	Head	Yuvaraj (room)

H.3 Distribution & social

Entity	Role	Leader
IMPAL	Managing Director	Mukund S. Raghavan (DIN 03411396, eff. 5 Jul 2025) (room)
IMPAL	Whole-Time Director	N. Krishnan
IMPAL	Chairman (non-executive)	S. Ram
Sundaram Motors	Executive Director	Sharath Vijayaraghavan (DIN 00243575) (room)
Sundaram Motors	EVP-Cars / Brand CEO	SS Ram / S.S. Ramasubramaniam (room)
Sundaram Motors	SGM-Finance	Suryanarayanan D (room)
Madras Auto Service	EVP / CEO	Gopal Maloo / Jacob (room; public naming a gap)
Sundaram Business Services	CEO	Deepa Ramesh (room)
Sundaram Medical Foundation	Director	Arjun Rangarajan (room)
Laxmi Charities	CEO	Shyamala Ashok (room)

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PART I — Function-by-Function AI Implications (Investments, Insurance, Shared Services & Social)

Completes the function-by-function mapping begun for lending (C3.7) and manufacturing (C1.8).

I.1 Investments & Insurance

Role cluster	Operating tension	Highest-confidence AI implication
AMC MD / CBO	Retail-led growth vs distributor-channel complexity; mix broadening post-Principal integration	Distributor-servicing intelligence, channel reconciliation, investor-onboarding assistance — within SEBI/CSCRF controls
AMC CFO / Chief Business Operations Officer	Daily NAV obligations; fund-accounting integration; reconciliation load	Break/anomaly detection and reconciliation assistance with strict accuracy and audit trails
AMC Senior Fund Manager	Research throughput; document/disclosure synthesis	Research and disclosure summarisation as decision-support, not autonomous allocation
Alternates MD / Fund Manager	Affluent/family-office expectations; PMS/AIF conflict management; closed-end administration	Onboarding/KYC/accreditation acceleration, capital-call/distribution reporting, suitability evidencing
Alternates Zonal Head	UHNI/family-office sales; bespoke reporting credibility	Client-reporting automation and relationship-manager productivity
Royal Sundaram MD / COO	Claims speed/consistency across siloed channels; phygital experience	Claims triage/orchestration and intake automation (ML+RPA already in use) under IRDAI guardrails
Royal Sundaram CFO	Combined ratio; underwriting discipline	Underwriting consistency support, fraud/leakage detection in claims
Royal Sundaram CDOs (Retail/Commercial)	Dual retail/commercial channel and portal sprawl; intermediary management	Channel-consistency tooling, agent enablement, multi-portal servicing harmonisation

I.2 Shared Services & Social

Role	Operating tension	Highest-confidence AI implication
SBS CEO (Deepa Ramesh)	Shift from T&M to outcome/AI-native delivery under strict client-data isolation; seasonal SMSF peaks	Agentic SMSF reconciliation and IDP for mortgage underwriting — executed inside client clouds, preserving isolation (potential ~80% manual-prep reduction)
SMF Director (Arjun Rangarajan)	Clinician documentation burden; limited outreach capacity; consent under DPDP	Ambient clinical scribing (35–45% documentation-time cut) and scaling the “Dear Akka” triage assistant — under ICMR ethical-AI guidelines and human oversight
Laxmi Charities CEO (Shyamala Ashok)	Reactive student support; volunteer-trainer constraints; minor-data consent vs CSR reporting	Predictive attrition flagging over the CMI database and gen-AI English tutoring for Vridhi — with verifiable parental consent and minimised identifiers
Group HR leaders (Wheels/Axles/SHF)	Disconnected manufacturing vs financial-services talent pools	Skills mapping, internal-mobility matching and HR-process automation across entities
Group digital mandates (CTDO/CIDO)	No group AI CoE; federated data ownership; responsible-AI gap	Shared AI/data backbone, cross-entity entity resolution, and a model-governance/consent architecture the others consume

PART J — Facilitator Notes per Table

What to confirm and probe at each table before and during ideation — discovery prompts, not solutions. Each pairs the table’s territory with the open questions and existing assets it must not duplicate.

- **T1 (Braking & Foundry):** Confirm foundry instrumentation/historian coverage at Padi/Naidupet; confirm whether casting-quality data is already linked to DISA logs; surface the real cost-of-quality and energy-tariff exposure. Do not re-propose DISA automation (already live).
 - **T2 (Wheels & Components):** Confirm OEM pull-rate data availability and current forecasting method; confirm robotic-weld PLC data accessibility; probe working-capital/inventory KPIs. Avoid re-proposing existing robotic seam-tracking.
 - **T3 (Powertrain):** Confirm Turbo Energy’s Siemens-ACM data can be reused; probe end-of-line test data lineage for unbalance root-cause; for ABI, scope the aerospace first-article documentation burden and NADCAP constraints.
 - **T4 (Aftermarket & Retail):** Lead with sector reality, not entity claims; pre-empt the Sundaram Motors/Honda confusion; confirm IMPAL B2B-platform status and SKU/demand data; treat MAS capability as net-new until confirmed; reference myTVS as comparator only.
 - **T5 (Lending & Credit):** Map against SF’s existing data lake / IDP / agentic pilots to avoid duplication; keep human-in-the-loop on credit decisions; probe the informal-income evidencing gap in the emerging-business unit; respect RBI Digital Lending data-localisation.
 - **T6 (Collections & Branch):** Build on the existing EWS framework rather than rebuilding it; confirm field-collections vs tele-calling vs legal workflow mix; respect recovery-agent conduct rules.
 - **T7 (Investments, Wealth & Insurance):** Distinguish Royal Sundaram’s live ML/RPA/claims estate from AMC/Alternates’ lighter AI footprint; keep NAV/claims accuracy and SEBI/IRDAI controls central; separate retail vs commercial insurance realities.
 - **T8 (Group Finance & Risk):** Acknowledge the core is data-integration; scope the AI-tractable slice (mapping/anomaly/narrative for close; transaction-matching for intercompany; cash forecasting for treasury) rather than a “single source of truth” platform.
 - **T9 (Group Enablement):** Treat cross-entity master data as an organisational + DPDP problem as much as a technical one; anchor on the CTDO/CIDO mandates; scope a thin entity-resolution slice; for HR keep it concrete; for shared services respect client-data isolation.
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PART K — Consolidated Source Register (key families per workstream)

WS	Primary sources	Type
WS0	MCA/ROC master data (ZaubaCorp/Tofler/Tracxn); NSE/BSE scrip pages & SF Reg.31(4) shareholding (2 Apr 2026); NHB HFC list; SEBI MF register; IRDAI Reg.102 docs; Madras HC judgments (SMF trust deed); M&A Critique / Industrial Economist / Business India (family settlement); Autocar Professional (Axles/Dana); MarkLines (BIADS/ADVICS)	Registry / exchange / regulator / court / press
WS1	Wheels India & listed-entity ARs/investor decks; CRISIL/ICRA/India Ratings/CARE rationales; ACMA, Autocar Professional, MotorIndia, Pro MFG Media; Brakes India sustainability report; ZaubaCorp masters; vendor case studies (Siemens ACM, DISA)	Primary financials / ratings / trade press / capability catalogues
WS2	Sundaram Finance FY25 AR & FY26 audited results; SF product/network pages & SF Next; Sundaram Home FY25/FY26 results, customer booklet, EB pages; RBI SBR framework & UL list, Digital Lending Directions 2025, KYC/outourcing/IT-governance directions, HFC Directions 2021; Times of India (SHF growth)	Listed-company primary / regulator / management-linked press
WS3	Sundaram AMC FY25 AR & site; Sundaram Alternates site & PMS disclosure document; Royal Sundaram FY25 AR, portals & press (FirstApex, IRCTC/Insillion, rebrand); SEBI MF/AIF/PMS master circulars & statistics, CSCRF; IRDAI products/policyholder/cyber regulations; AMFI March 2025 note	Statutory AR / regulator / official press
WS4	IMPAL exchange filings/AR & Tickertape; Mercedes-Benz/Sundaram Motors brand pages; mastvs.com (MAS); ACMA FY25 release, ICRA aftermarket outlook, FICCI counterfeit data; CARS24 used-car report; NADA/Edmunds dealer economics; GoMechanic/Boodmo/Spinny coverage; myTVS & Toyota-Kirloskar JV press	Listed filings / OEM & entity sites / sector bodies / trade press
WS5	SBS/SMF/Laxmi websites & brochures; Sundaram Finance AR & Brakes India sustainability report; MeitY DPDP Act 2023 & Rules (Nov 2025); MCA Form CSR-1 notifications; ABDM/SAHI/BODH strategy docs; ICMR ethical-AI guidelines; NASSCOM-EY AI Adoption Index; Bain APAC healthcare report; EY/Deloitte BPM reviews	Primary entity / government / sector research
WS6	TSF Group & entity sites; Sundaram Finance & Royal Sundaram ARs and governance/board materials; IDC event page (CIDO); RBI Digital Lending & IT-governance directions; IRDAI cyber circular; SEBI CSCRF; entity careers pages	Corporate governance / official sites / regulator

PART L — Consolidated Inference & Assumption Log

Claim	Confidence	Basis
In-scope cluster = T.S. Santhanam branch (TSF Group)	Verified	Multi-source press synthesis + MCA
Family allocation of Wheels/Brakes/IMPAL/Sundaram Motors/MAS to TSSFPL	Inferred	Business press; not a primary filing
SF promoter group 37.21% (Apr 2026), dispersed across ~84 individuals/HUFs + ~10 holding cos	Verified	SEBI Reg.31(4) disclosure to NSE
Axles India → 62.98% SFHL (2025), Dana 48.33% exit	Verified	Autocar Professional citing NSE filing
SF NBFC-ML; SHF HFC-ML	Inferred (high)	RBI SBR rules + absence from Jan-2025 UL list
SHF FY25 deposit base ~₹2,345 crore	Inferred	Back-solved from top-20 deposit concentration
“Others” (9.44%) in SF asset mix ≈ SME/leasing/fuel/tyre/NBFC finance	Inferred (medium)	Product menu vs standalone asset-class table
Underwriting at both lenders remains materially judgment-led	Verified	Collateral-led structures, RCU/EWS, no public automated scorecard
Biggest lending AI value = augmentation, not autonomous approval	Verified	Branch-heavy network, relationship positioning, regulation, strong asset quality
AMC operational pain concentrated in valuation/NAV/reconciliation	Inferred (medium)	Broad shelf, daily NAV, large distributor count, fund-accounting integration
Royal Sundaram operates a real retail/commercial distribution split	Verified	Separate retail/agent/commercial portals
“AI in production at AMC”	Unverified	No explicit public deployment claim found
Manufacturing pain magnitudes (yield, tool-life, drift)	Inferred (where entity-specific data thin)	Sector norms applied; foundry-yield band evidenced
Build tractability highest in manufacturing	Inferred (medium)	Less RBI/IRDAI/SEBI data encumbrance vs strong AI ambition in finance
CIDO remit bridges enterprise IT + manufacturing OT	Inferred	Title verified; remit not in public disclosure
Flometallic India as a potential in-scope entity	Unverified	Brakes India co-shareholder; confirm live
MAS leadership; Sundaram Motors EVP-Cars/SGM-Finance names	Assumed/Gap	Not in public sources; confirm at immersion
“Sundaram Honda” / “TVS Mobility OES & Lubes”	Excluded	Rajam branch, out of scope

PART M — Workstream Detail Appendix (folded-in detail from WS1-WS6)

This appendix folds in the detailed material from the seven source reports that the main body summarised or compressed — the per-process regulatory maps, lifecycle pain maps, full leadership rosters, do-not-propose lists, milestone timelines, evidence grids and benchmark specifics. It supplements (does not replace) Parts B-L; where a figure here is more granular than the body, this appendix governs.

M1 — Manufacturing & Auto Components (WS1)

M1.1 Quality Control as a distinct pain stage

Beyond the foundry/machining/assembly stages in C1.3, the cluster has a distinct **quality-control pain**: manual visual inspection of complex cast surfaces (e.g. steering knuckles, calipers at Brakes India) suffers defect-detection decay after consecutive shift hours; and although automated end-of-line testing (eddy-current, X-ray, 3D optical/laser scanning) is deployed at Turbo Energy and ABI-Showatech, these run as **disconnected workstations** — quality data is not fed back upstream, so there is no real-time closed-loop control to prevent defect recurrence. This is the technical root beneath challenge T1-S1.

M1.2 Function-by-function for the smaller entities

- **ABI-Showatech (MD Ramanujam Srikanth; ED Sundar K)**: automotive→aerospace diversification needs AS9100 conformance — AI correlating casting-mould variables with NADCAP NDT results to automate compliance tracking and accelerate aerospace first-article qualification (root beneath T3-S1).
- **Sundaram Composites / Leggero (CEO Rajiv Balagangadharan; COO G. Thirumalai)**: ML over autoclave temperature-pressure-time profiles vs composite void-ratio to optimise high-value cure cycles.
- **BIADS (Head of Greenfield Assembly)**: the Shoolagiri launch (Mar 2026) is a greenfield chance to embed native IoT connectivity and standard data models into mechatronic lines from day one.
- **Sundaram Dynacast (Head of Operations)**: AI predictive die-wear models to schedule tooling maintenance on actual shot degradation and prevent flash defects.

M2 — Lending (WS2)

M2.1 Full named leadership (supersedes the role-only entries in Part H)

Sundaram Finance: Harsha Viji (Executive Vice Chairman); Rajiv C. Lochan (MD); A.N. Raju (Deputy MD); **M. Ramaswamy (CFO)**; **Mukund Raghavan (Chief Risk Officer — distinct from the IMPAL MD of the same name)**; **P.N. Srikant (Company Secretary / Compliance Officer)**; Sudheer Warriar (CTDO); **Rajmohan Vasudeva Pai (Chief Credit Officer)**; plus named heads of Distribution, HO Operations, Collections & Recoveries, Deposits, **CPU, Analytics, Digital Initiatives**, and the regional heads (TN, North, West, MP/Chhattisgarh & East, Karnataka, Andhra, Kerala, Rajasthan, Chennai Zone, Salem & Dharmapuri Zone). **Sundaram Home Finance: Lakshminarayanan Duraiswamy (MD)** (the “Laks” used elsewhere); **P. Nishant (CBO)**; **Hema Ganapathy (COO)**; **P.S. Sridhar (CFO)**; **Ganapathi Subramanian (CHRO)**; **R. Mugunda Krishnan (CTO)**; **Kumaresh Balasubramaniam (Projects Head)**.

M2.2 Product specifics beyond term lending

Fuel / diesel finance is a low-cost, **card-based working-capital** facility usable at major fuel networks, with daily limit controls, online visibility and **up to 45 days’ credit** — generating card-control, merchant-reconciliation, daily-limit-exception and usage-anomaly process pains distinct from term lending. **Tyre finance** repays in **4 / 6 / 8 instalments** and serves fleet operators, manufacturers, dealers and CV/CE/school-bus owners. Emerging-business APRs at SHF are materially higher than prime, consistent with thinner files and manual cash-flow assessment.

M2.3 Emerging-business growth detail

SHF emerging-business disbursements **more than doubled to ₹590 crore in FY26 from ₹229 crore in FY25** — strategically important but still not the dominant book (a scaling frontier, not the current centre of gravity).

M2.4 Lifecycle pain map

Stage	Documented friction
Origination	Heterogeneity, not just volume — custom income assessment, seasonal repayment structures, asset-specific offerings imply field interpretation of cash flow rather than salary-slip processing; SHF EB explicitly targets self-employed/informal/documentation-challenged borrowers
KYC & documents	A real control surface — SHF was penalised by RBI in 2024 for periodic-KYC-updation failures; multiple digital touchpoints create cross-channel identity/document-version/service-consistency issues
Credit appraisal	In-house legal/technical teams + RCU screening; management obtains additional collateral on evaluation — a high-human-review process with selective analytical augmentation, not a low-touch scorecard
Disbursal	Direction is “faster/STP,” but by-product/by-region sanction-to-disbursal times, exception rates, rework rates and defect reasons are not in the public record — open questions for the room
Servicing	Multi-channel self-service already live (portals, quick-pay, e-NACH, QR) — reconciliation, next-best-action, complaint resolution and assisted self-service are immediate AI-relevant areas
Collections	Complaints cluster in loan-account/sanction-disbursal/legal categories; recovery-agent conduct rules bind outsourced field recovery — AI’s role is prioritisation, triage, field-intelligence capture and legal-pack readiness, not replacing judgment

M2.5 Guardrail-to-lifecycle map

Lifecycle stage	Principal guardrails	Practical implication
Lead generation / channel onboarding	Digital Lending Directions 2025; outsourcing accountability; fair-practice	DSA/DMA/LSP sourcing needs traceable consent, fee transparency, partner due diligence, clear RE accountability
KYC & onboarding	KYC Direction 2016; IT-governance & cyber controls	KYC refresh, document validity, beneficial-ownership, secure handling — SHF’s 2024 penalty shows real supervisory consequence
Credit appraisal & underwriting	SBR governance; internal controls; HFC Directions 2021; fair-practice code	AI credit tools need human accountability, auditability, documented override logic; autonomous approval is not current reality
Sanction & disbursal	Digital Lending KFS/disclosures (where digital); HFC MITC/fees; outsourcing/IT	Sanction comms, fees, APR, prepayment, e-mandates must be exact and channel-consistent
Servicing & self-service	KYC; grievance redress; IT governance; cyber	Portals/apps need strong identity, change-management, audit trails, complaint routing
Collections & recovery	RBI recovery-agent circular; fair-practice/contact rules	Calling discipline, field conduct, repossession/legal workflow, partner oversight — AI prioritises, never weakens conduct controls
Portfolio monitoring & EWS	SBR risk-management; HFC prudential; cyber/data	Early-warning tools need versioning, monitoring, board-visible governance as SHF expands into thinner files
Model, data & vendor governance	IT-governance & IT-outsourcing directions; digital-lending accountability	The entity remains responsible even where models/software are vendor-supported; a dedicated AI-governance framework is an open agenda

The governing principle: **regulation does not prohibit AI in these lenders — it prohibits unaccountable AI**. Any agenda that increases opacity in field discretion or channel accountability runs against the regulatory grain.

M2.6 Built-versus-not-evidenced capability matrix

Capability	Sundaram Finance	Sundaram Home Finance	Status
Digital customer servicing	Vehicle/fuel/deposit portal; digital deposits; SF Next	Deposit portal, BOS logins, quick pay, QR, e-NACH, document upload	Built
Workflow digitisation	Guided workflows, task management, STP, communication framework	BOS + branch/customer/partner workflows	Built / scaling
Data platform	Cloud data lake for Systems of Intelligence	Not publicly evidenced	Built at SF; unverified at SHF
ML / AI	More ML models; agentic pilots	No explicit public AI claim	Built/testing at SF; not claimed at SHF
Fraud / document analytics	Not explicitly named	RCU screening + EWS framework	Built
Cyber & IT control	24x7 SOC, VAPT, external validation	Cyber-risk function; CTO/CRO roles	Built
Chatbots / branded RPA / propensity stack	No reliable public evidence	No reliable public evidence	Gap

M3 — Investments & Insurance (WS3)

M3.1 Sundaram Alternates — named strategies

Publicly marketed PMS strategies: **FIRST (Fixed Income Rated Short Term), Rising Stars, Voyager, SELF, SISOP**. The registered strategy universe in the PMS disclosure document is larger — including **SUNOPP, SUNIOP, SITP, Quant, India Equity, India Dedicated, Wonder Companies, Strategic Equity, Alpha Equity** and FIRST plan variants — so the marketed shelf is only a subset of the registered architecture.

M3.2 Royal Sundaram — additional scale figures

FY25 **PAT ₹160 crore; 145 branches** (annual report, as at 31 Mar 2025) versus 135+ on the public site; the site markets **4 crore+ customers** and **2,000+ employees; total motor-insurance premium ₹1,720.67 crore** (distinct from the commercial-motor ₹1,414 crore already noted).

M3.3 Operational pain map by sub-business

- **AMC:** valuation/NAV computation/fund accounting (multi-category: debt, hybrid, index, FoF, each with differing valuation/disclosure burdens); cash/unit/channel reconciliation across a very large distributor base and multiple utility rails; and **change-control/operating-model integration** following the **FY24 demerger of the fund-accounting division from Sundaram Fund Services into the AMC**.
- **Alternates:** investor onboarding/KYC/suitability (PMS minimum-investment thresholds, risk profiling, separate books); closed-end **capital-call/distribution/reporting/lifecycle administration**; and **PMS→AIF conflict management**, with reliance on Sundaram AMC for research, trading, IT, risk, marketing and administration (efficient but control-intensive).
- **Insurer:** underwriting/product-governance consistency across personal/commercial/commercial-motor; **claims orchestration** across high-volume omnichannel intake/networks; and distribution/intermediary management across separate retail/commercial/partner ecosystems.

M3.4 SEBI guardrails by process stage (AMC and AIF/PMS)

Process stage	AMC guardrail	AIF/PMS guardrail	AI implication
Product design & launch	MF master circular; MF Regulations/categorisation	AIF master circular; PMS Regulations + master circular	Model-led product changes still need the same filings, disclosures, approvals
Investor onboarding & suitability	KYC/PAN/nomination/distributor conduct; riskometer	PMS risk profiling, ₹50 lakh minimum for non-accredited investors , client agreements, pre-account disclosure; AIF PPMs for sophisticated investors	AI onboarding must preserve suitability evidence, consent traceability, reviewability

Portfolio management & dealing	Valuation, riskometer, expense, disclosure rules	PMS fiduciary duties, segregation of books; AIF fund-level governance/valuation/PPM	AI in research/dealing/rebalancing sits under human accountability and conflict-management
Reporting & disclosures	Expense/return/yield/riskometer disclosure	PMS performance/fee/conflict/grievance disclosure; AIF PPM architecture	AI-generated reports cannot relax mandated content, cadence or auditability
Cyber & operational resilience	SEBI CSCRF	Same expectations as SEBI-regulated intermediaries	AI raises model-security, data-governance, third-party-risk obligations

M3.5 IRDAI guardrails by process stage (insurer)

Process stage	Primary IRDAI guardrail	AI implication
Product design & filing	Insurance Products Regulations 2024; Master Circular on General Insurance Business	AI pricing/design subordinate to board-approved product governance and file-and-use
Sales / distribution conduct	Protection of Policyholders’ Interests master circular 2024; intermediary/commission rules	AI sales must not create undisclosed inducement, unsuitable selling or opaque communication
Operations & servicing	Master Circular on Operations & Allied Matters; Corporate Governance for Insurers 2024	AI workflow automation sits inside management accountability and board-governed controls
Claims handling & grievance	Protection of Policyholders’ Interests master circular 2024	AI triage cannot displace statutory claims-handling/grievance obligations
Data retention, sharing & cyber	Maintenance of Information Regulations 2025; Information & Cyber Security Guidelines 2026	AI/data platforms raise data-lineage, vendor-risk, access-control, model-security duties

M4 — Distribution & Aftermarket (WS4)

M4.1 IMPAL milestone detail

FY23-24: opened four new branches, added a new product line, **PBT ₹89.16 crore**; **Jan 2024 interim dividend ₹9/share**; FY24-25 launched the B2B dealer-ordering platform and **amalgamated CAPL Motor Parts** (network → ~89 branches / 20 states / 3 UTs); 16 May 2025 board approved the leadership change; **5 Jul 2025** Mukund S. Raghavan became MD (N. Krishnan → Whole-Time Director); **23 Jan 2026** Q3/9M FY26 results with **₹10 interim dividend**.

M4.2 Benchmark specifics

Player	Type	Notable signal
myTVS / Ki Mobility (Rajam, out of scope)	Multi-brand platform	Cloud-AI predictive service; Drivenostics diagnostics; Life360 super-app; dark-store quick-commerce; ~1.2 cr-SKU catalogue; ~10M customers; ~₹2,100 cr FY26; EBITDA-positive
GoMechanic → Spinny	Service aggregator	Collapsed on fabricated financials (Jan 2023); acquired by Servizzy then Spinny (Dec 2025)
Boodmo	Parts e-commerce	Multi-vendor catalogue with VIN/vehicle search , pan-India logistics; tens of thousands of orders/month
Cars24 / Spinny / CarTrade	Used-car platforms	AI valuation, predictive maintenance, embedded lending; Cars24 ~₹500 cr tech spend (Dec 2024)
DMS vendors (e.g. Autoboom)	Dealer software	GST-ready invoicing, service reminders, inventory, CRM; cited ~40% service-efficiency gains

M4.3 Function-by-function (distribution roles)

- **IMPAL MD (Mukund S. Raghavan):** demand-forecasting/inventory optimisation across ~89 branches × ~18,000 dealers; working-capital/receivables visibility; counterfeit-counteracting provenance; extending the B2B platform; EV portfolio watch.
- **Sundaram Motors ED (Sharath Vijayraghavan):** profit pool is fixed operations + retention — service throughput, CX/CRM, CPO economics under 18% GST; leverage the densest southern Mercedes network for capacity utilisation.
- **Sundaram Motors EVP-Cars:** omnichannel lead management, test-drive/booking digitisation, used-car trade-in integration for a low-volume high-value brand.
- **Sundaram Motors SGM-Finance:** working-capital intensity, floor-plan/inventory financing, receivables, fixed-ops gross-profit and parts/labour-margin discipline.
- **MAS EVP/CEO:** availability, counterfeit competition, demand planning and a near-absent digital footprint — the thinnest-documented entity, where discovery is most valuable.

M4.4 Evidence grid (known / partial / unknown)

Theme	IMPAL	Sundaram Motors	MAS
Legal identity & ownership	Known	Known	Known
Brands/franchises	Known	Known (Mercedes)	Partial
Footprint/outlet counts	Known (~89)	Known (~9–10)	Partial (~75)
Financial scale	Known	Unknown (private)	Unknown (private)
Leadership	Known (MD)	Known (ED, Brand CEO)	Unknown
Customer / operational pains	Partial	Partial / Unknown	Unknown
Digital/data maturity	Partial (B2B platform)	Partial (OEM DMS)	Unknown

Existing AI/automation	Minimal/none	Minimal (OEM)	None found
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M4.5 Distribution-specific open questions

1. IMPAL B2B platform — adoption across 18,000 dealers, functionality, any demand/inventory layer? 2. Inventory planning across ~89 branches — centralised vs branch, fill rates, stock-turn? 3. Managing ~60% CV-spares + EV-transition portfolio risk? 4. Working-capital/receivables process across the dealer network? 5. Sundaram Motors DMS (SKYLine/Zoho) and customer-facing digital beyond OEM mandates? 6. Service-bay throughput, technician capacity, CSI; CPO materiality post-GST? 7. Confirm the in-scope perimeter (TSSFPL vs TVS Mobility) today? 8. MAS leadership, exact branch count/geography, product lines, systems? 9. MAS vs Rajam-side OES & Lubes delineation? 10. How are genuine parts authenticated today? 11. What AI/digital is already in flight (do-not-duplicate)? 12. What data exists (sales/inventory/service/customer) and in what quality?

M4.6 Do-not-propose (distribution)

IMPAL B2B dealer-ordering platform (confirm scope/adoption); IMPAL EV-component diversification; Sundaram Motors Mercedes **“SKYLine” DMS (Zoho)** and OEM-level online booking; and — adjacent, not to attribute to in-scope entities — myTVS Life360 / Drivenostics / dark stores and the TSSF-Toyota-Kirloskar “TSERV Select” and “UTrust” JVs.

M5 — Shared Services & Social (WS5)

M5.1 Executive action map

Leader	Core process	Targeted AI opportunity	Critical data constraint	Action item
SBS CEO	SMSF administration & portfolio accounting	Agentic multi-agent LLM to automate bank reconciliations and draft audit workpapers	Client-DB isolation under ISO 27001/GDPR/APP	Pilot agentic document processing in the SMSF division
SBS CEO	Mortgage broker/aggregator support	IDP to verify payslips and tax filings	Australian lending regs; strict NDAs	Map document workflows; identify automation areas
SMF Director	Clinical documentation & consults	Ambient voice-scribing to draft structured EHR notes	Prior patient consent; secure storage	Sandbox ambient scribing in low-risk OP clinics
SMF Director	“Dear Akka” triage & outreach	WhatsApp conversational assistant	Clinical-protocol alignment; explicit consent	Validate chatbot logic with gynaecology specialists
Laxmi CEO	Scholarship nominations & partner coordination	Automate screening; streamline nominations	Verifiable parental consent for minors	Implement consent gateways in the application portal
Laxmi CEO	Student tracking & retention	Predictive analytics on the CMI database	Protect grades/attendance; deletion processes	Build early-outreach predictive models with CMI

M5.2 Milestone timeline

1973 Laxmi Charities founded · 1990 SMF registered · 1994 SMF hospital operations begin · 1998 annual FORUM CME conferences · 2001 SBS incorporated; SMF DNB training begins · 2004 SBS enters the Australian SMSF market · 2023 DPDP Act enacted · 2025 revised Form CSR-1 enforced; DPDP Rules notified · 2026 “Dear Akka” digital companion active; SAHI national framework released.

M5.3 Additional entity facts

SMF is **NBE-accredited** and runs the **STEP** (Support Therapy to Educate People) outreach initiative; **Laxmi Charities partners 36 schools and colleges** and maintains the CMI student-tracking database.

M5.4 Do-not-propose (shared & social)

- Consumer medical-diagnostic AI **without a clinician in the loop** (SAHI mandates AI augments, not replaces, clinical judgment).
- **Cross-entity patient→donor/beneficiary database integration** (violates DPDP purpose-limitation/minimisation/consent; severe penalties).
- AI directed at the **two-wheeler branch** (out of scope).
- **Fully automated AI scholarship disbursal** without human oversight (right to object to significant automated decisions).

M5.5 Open questions & named assets

Open questions: SBS T&M→outcome-based contract transition; a centralised group DPO framework across BPO/healthcare/philanthropy; “Dear Akka” clinical/technical validation; Laxmi verified-parental-consent redesign; reconciling CSR beneficiary-metric demands with DPDP minimisation. *Named assets:* “Dear Akka” women’s-health companion (SMF + CAN-STOP, 7 conditions, Tamil/English); Vridhi English initiative; CMI student-tracking database; SMF Digital Health Visual Division; SBS Learning Management Systems.

M6 — Group AI Posture (WS6)

M6.1 The full dual ranking

Strongest disclosed AI/digital capability: 1) Lending → 2) Investments & insurance → 3) Manufacturing → 4) Distribution → 5) Shared & social. **Most favourable thin-slice build surface after weighting regulation:** 1) Manufacturing (bounded pilots, low consumer-data friction, strong automation/traceability substrate) → 2) Lending (best AI estate but tightly RBI/DPDP-bounded) → 3) Investments & insurance (good estate, IRDAI/SEBI + JV boundaries) → 4) Distribution (channel opportunity, weak data foundation) → 5) Shared & social (too little reusable-platform evidence). This dual ranking is what resolves the apparent contradiction: **AI ambition looks finance-led; first-build feasibility may be manufacturing-led.**

M6.2 Soft constraints (organisational)

- **Separate legal entities, not a unified enterprise:** notably, **Sundaram Finance Holdings ceased to be a subsidiary during the quarter ended March 2024** (yet some pages still list it as one); Royal Sundaram is a JV with Ageas; no public architecture shows how data can legally/contractually/operationally move across entities.
- **Relationship-centric intentionality:** SF’s “high touch + high tech” is a declared **constraint on over-automation** — proposals that

- disintermediate RMs/branches/human judgment without clear need run against public self-definition.
- **Information fragmentation:** divergent group metrics/branch counts across official pages indicate decentralised stewardship, which maps to uneven master-data governance and function-led (not enterprise-led) builds — making scale and reuse harder.

M6.3 Function-by-function and the “light group spine”

- **Brakes India CIDO:** convert a strong automation/traceability substrate into a legible AI roadmap (visual-inspection assistance, quality root-cause copilots, maintenance diagnostics, line-change support, engineering-knowledge retrieval anchored to AMTIX) — without overclaiming enterprise-AI maturity.
- **Sundaram Finance CTDO:** the question is not “start AI” but “govern AI from a better base” — which use cases stay within the high-touch model, how model-governance catches up with experimentation, and how much reuse is safe across adjacent financial entities.
- **Group leadership:** rather than a premature central CoE, define a **light group layer** — shared principles, common vendor/risk review, secure knowledge-sharing patterns, a reference architecture, and a do-not-duplicate list. There is no public group spine today.

M6.4 WS6 limitation and AI-operating-model open questions

WS6 was run **without access to WS1-WS3 outputs**, so its reconciliation is a provisional public-signals view (manufacturing = strong digital substrate / thin AI articulation; lending = clearest AI momentum; investments & insurance = Royal strong, AMC AI-light). Its highest-value unresolved questions are operating-model specifics: the **CIDO’s reporting line and budgeted AI roadmap; where Sundaram Finance’s ML models actually sit** (pricing, risk, collections, fraud, service or workflow); whether a **shared cross-entity data architecture** exists beyond related-party service arrangements; and whether **Royal Sundaram’s AI governance** is embedded privately in underwriting/claims controls.

M6.5 Consolidated do-not-propose (group AI estate)

Treat as already built or in motion — do not re-pitch: SF cloud data lake / Systems-of-Intelligence foundation; SF ML models; SF agentic pilots; SF intelligent document processing; SF 24×7 SOC + CISO operations; Royal Sundaram AI/digitisation pilots; Royal Sundaram ML + RPA in operations/claims; Royal Sundaram core-platform modernisation on private cloud (FirstGen v7 / NIP v3 / FirstApex); Brakes India AMTIX machine-building and configurable automation; Brakes India cloud-linked traceability and robotic factory automation; Sundaram Mutual Investor App and InvWise distributor platform.

End of Part M. With this appendix folded in, the consolidated pack now carries the substantive detail from all seven workstreams — topology, AI posture, the five domain deep-dives, clustering, challenge portfolio, build-readiness, and the previously-omitted per-process regulatory maps, lifecycle and sub-business pain maps, full leadership rosters, milestone timelines, evidence grids, benchmark specifics and do-not-propose lists.

PART N — Complete Remaining Detail, Open Questions & Source Registers (WS1-WS6)

This part folds in every remaining item from the six domain workstreams that the body and Part M had not yet captured: residual figures, the phased strategic interpretations, each report’s own open-questions and do-not-propose lists, the full named-asset inventories, milestone timelines, evidence maps, and the source registers (reproduced in cleaned form — broken citation tokens and raw link placeholders removed; the verbatim numbered URL lists remain in the original reports).

N1 — Manufacturing (WS1)

N1.1 Residual financial figures

- **Wheels India:** FY26 net profit **₹139 crore (+31%)** (on ~₹5,098 crore standalone revenue); debt-to-OPBITDA **3.4x** (9M FY26); capex ₹275–300 crore.
- **Turbo Energy:** FY25 net profit **₹275.4 crore** (on ~₹2,960.5 crore operating revenue); free cash ₹986.1 crore; 86.4% diesel concentration; 67% domestic share.
- **Axles India:** FY25 **EBITDA margin 11.9%**; **top-5 customer concentration 89.0%**; Dana stake bought out by TSF Investments May 2025; 400,000 housings/yr.
- **Brakes India:** FY25 op. margin 13.0% (from 14.7% FY24); **Deming-Award foundry; 200,000 t iron-casting capacity; Dunes Oman 51%.**
- **ABI-Showatech:** FY25 operating income ₹554.8 crore (GDC + AS9100 aerospace divisions). **Sundaram Dynacast:** 68,000 sq ft Ambattur; ₹150–200 crore.

N1.2 Phased “Built → Scaling → Testing” interpretation

- **Phase 1 — Built (proven ROI):** scale Siemens **ACM** adaptive CNC control across high-precision lines at TEPL and ABI; standardise data pipelines feeding spindle-load/feed data into central historians for tool-wear modelling.
- **Phase 2 — Scaling (proven in pockets):** scale the BIPL-DISA automated sand-prep/laser-pouring systems across all group foundries; unify real-time pouring/moulding parameter databases on a shared dashboard for yield.
- **Phase 3 — Testing → Scale:** scale **computer-vision quality inspection** to replace manual visual audits on high-volume brake rotors/casting surfaces, with edge inference integrated to sorting actuators for automated reject handling.

N1.3 Strategic tensions / gaps

Systems-integration gap (no automated SCADA↔S/4HANA connectivity — analysis limited to manual batch exports); the **bullwhip** vulnerability of manual demand forecasting to abrupt OEM schedule changes; and the **EV talent squeeze** (mechanical designers must upskill in embedded software and electronic validation).

N1.4 WS1 open questions for the immersion

1. Current data latency exporting shopfloor variables (sand moisture, pouring temperature) to BIPL’s ERP — low enough for real-time predictive analytics?
2. Can Sundaram Dynacast’s proprietary multi-slide control cards export high-frequency sensor data without costly retrofits?
3. Barriers (incl. **BorgWarner JV/IP data-sharing restrictions**) to a shared BIPL-foundry ↔ TEPL-machining data channel?
4. Can BIADS Shoolagiri mechatronic-validation models be standardised onto BIPL’s legacy light-vehicle lines?
5. Can Wheels India’s ERP ingest high-frequency multi-facility edge streams for predictive maintenance?

N1.5 WS1 evidence map (entity → finding → source → status)

Entity	Finding	Primary source	Status
BIPL	Rev ₹7,509 cr FY25; op. margin 13.0%; capex ₹500–600 cr	CRISIL rationale, Jun 2025	Verified
BIPL	Deming foundry; 200,000 t capacity; Dunes Oman 51%	Corporate disclosures FY24	Verified
WIL	Rev ₹5,098 cr FY26; net profit ₹139 cr (+31%)	Standalone release, May 2026	Verified
WIL	Debt/OPBITDA 3.4x 9M FY26; capex ₹275–300 cr	ICRA rationale, Feb 2026	Verified
TEPL	Rev ₹2,960.5 cr; net profit ₹275.4 cr; free cash ₹986.1 cr	ICRA rationale, Oct 2025	Verified
TEPL	86.4% diesel; 67% share; sole-supplier status	ICRA Sep 2024 / MotorIndia Jul 2025	Verified
TEPL	Siemens ACM: 6–12% cycle-time cut; 487 hrs saved; 20% tool-cost cut	Siemens case study	Verified
AIL	400,000 housings; Sriperumbudur/Cheyyar/Jamshedpur	Corporate profile 2025	Verified
AIL	EBITDA 11.9% FY25; top-5 concentration 89.0%; Dana buyout May 2025	India Ratings Nov 2025	Verified
ABI	Op. income ₹554.8 cr FY25; GDC + AS9100 aerospace	ICRA May 2025	Verified
Leggero	MIND S.r.l. JV; Chennai plant Dec 2022	Partnership announcement	Verified
BIADS	Slip-control JV; Shoolagiri Mar 2026; ₹275 cr paid-up	MCA Nov 2025 / Pro MFG Mar 2026	Verified
Dynacast	68,000 sq ft Ambattur; multi-slide zinc; ₹150–200 cr	Form Technologies / Tofler	Verified

N1.6 WS1 source families

Tier-1/2: listed-entity AR/investor decks (Wheels India), CRISIL/ICRA/India Ratings/CARE rationales. Tier-3: ACMA, Autocar Professional, MotorIndia, Pro MFG Media. Tier-4: technical datasheets, Siemens/DISA case studies, recruitment portals. (Original carries ~81 numbered web citations incl. Zaubacorp masters, axlesindia.com, brakesindia.com sustainability report.)

N2 — Lending (WS2)

N2.1 WS2 do-not-propose screening list

Screen out as generic “new” ideas (already evidenced; only enhancement/re-architecture/control/scale-up is non-duplicative): SF customer self-service (vehicle/fuel/deposits); digital deposit opening/servicing; SHF online business origination; SHF channel-partner digital workflow; document upload; e-NACH; quick pay & QR; SF cloud data lake; SF ML models; SF agentic pilots; guided workflows & task management; SHF RCU underwriting screening; SHF EWS monitoring; SF SOC-led cyber operations.

N2.2 WS2 named assets and programmes

SF One customer portal; **SF Next** app; cloud data lake / Systems of Intelligence; ML models; agentic pilots; guided workflows / task management / communication framework / nano learning. **SHF**: BOS / home-loan online login; **eKYC NXTGEN**; e-NACH registration; document upload; RCU; EWS framework; **Anugraha / Anugraha-LAP / Nidhi** emerging-business suite.

N2.3 WS2 open questions for the immersion

Actual channel mix by product/region; branch-vs-partner TAT split; underwriting rework and exception rates; informal-income verification methods used today; override frequency in credit decisions; collections operating split (tele-calling / field / legal); digital-adoption share in servicing; deposit unit economics; and the real internal model inventory beyond “more ML models.”

N2.4 WS2 milestone timeline

2021 RBI scale-based regulation + HFC Directions framework · 2022 RBI recovery-agent outsourcing circular; SF CTDO employment begins (ex-TCS) · 2023 RBI IT-governance & IT-outsourcing directions finalised · 2024 SHF RBI penalty for KYC periodic-updation non-compliance · FY25 SF discloses data lake / more ML models / agentic pilots; SHF EB workforce & branch expansion · Jan 2025 RBI Upper Layer list excludes SF and SHF · May 2025 RBI Digital Lending Directions 2025 · FY26 SF net stage-3 0.69%; SHF gross stage-III 1.11% / net 0.51%; SHF AUM & EB growth accelerate.

N2.5 WS2 evidence map

Theme	Strongest evidence	Confidence
SF core book is vehicle/equipment-led	FY25 standalone asset-class concentration	High
SHF broader than plain home loans	Customer booklet / product pages	High
Branch remains central	SF 600 / SHF 115 / EB 72	High
SF has real AI ambition underway	Data lake, ML, agentic pilots, STP	High
SHF targeting informal-income borrowers	EB page language	High
Digitally augmenting, not replacing field	Portal estate + high-touch/high-tech	High
Asset quality strong	FY26 stage ratios	High
Regulation shapes AI design materially	SBR, DL 2025, KYC, outsourcing, HFC	High
Channel mix / true TAT / scorecard share / line-level credit cost	Not disclosed	High-confidence gap

N2.6 WS2 source register (cleaned)

Title	Publisher	Type	Confidence
FY25 Annual Report	Sundaram Finance	Primary	High
FY26 audited results	Sundaram Finance	Primary	High
Product pages (CV/car/CE/SME/leasing/fuel/tyre/farm)	Sundaram Finance	Primary	High
Network page	Sundaram Finance	Primary	High
FY25 Annual Report	Sundaram Home Finance	Primary	High
FY26 audited results	Sundaram Home Finance	Primary	High
Customer booklet / product pages	Sundaram Home Finance	Primary	High
About / emerging-business pages	Sundaram Home Finance	Primary	High
Scale-Based Regulation framework + Upper Layer list	RBI	Regulator	High
Digital Lending Directions 2025	RBI	Regulator	High
KYC Direction	RBI	Regulator	High
Recovery-agent / outsourcing circular	RBI	Regulator	High
IT-governance & IT-outsourcing directions	RBI	Regulator	High
HFC Directions 2021	RBI	Regulator	High
SHF FY26 growth coverage	Times of India	Secondary	Medium

N2.7 WS2 inference log

SF in RBI Middle Layer (deposit-taking NBFC, absent from UL list) — High; SHF in Middle Layer (HFC, absent from UL) — High; SF “others” bucket ≈ SME/leasing/fuel-tyre/NBFC finance — Medium; SHF FY25 deposit book ~₹2,345 cr (back-solved) — Medium; underwriting materially judgment-led at both — High; biggest AI value = operational augmentation, not autonomous approval — High.

N3 — Investments & Insurance (WS3)

N3.1 Corporate event previously absent

2021 — Sundaram AMC announced the acquisition of Principal AMC in India, adding large- and mid-cap capability to a franchise traditionally weighted toward mid/small-cap.

N3.2 Sundaram Alternates fund-level detail & roadmap

- **RE Credit Fund IV** closed with **₹1,217 crore** commitments (FY25).
- **Performing Credit Opportunities Fund Series I** launched, expected to close the next fiscal after initial close.
- Other named vehicles: **Emerging Corporate Credit Opportunities Fund I; ATLAS; ATLAS II; ACORN; Sundaram India Premier Fund.**
- Roadmap read: continued private-credit scaling and selective equity-platform expansion; not-claimed = any large public AI-led operating transformation.

N3.3 WS3 milestone timeline

1996 Sundaram AMC / Sundaram Mutual founded (under SF sponsorship) · 2000 Royal Sundaram IRDAI licence (Oct) · 2009–2011 Alternates launches Rising Stars, SISOP, SELF, Voyager PMS strategies · 2018 Sundaram Alternate Assets spun off from Sundaram AMC (SEBI-approved AIF manager transition) · Jan 2019 Ageas strategic stake in Royal Sundaram; Alternates PMS operations effective 2 Jan 2019 · 2021 Sundaram AMC Principal-AMC acquisition · 2023 Alternates RE Credit Fund IV launched · FY24 fund-accounting demerger (Sundaram Fund Services → AMC) · Feb 2024 Anand Radhakrishnan CEO; Jun 2024 assumes MD · FY25 Business Cycle Fund launched; RE Credit Fund IV closed ₹1,217 cr; Performing Credit Opportunities Fund Series I launched · May 2025 Vedanarayanan Seshadri MD (eff. 28 May) · Sep 2025 Royal FirstGen v7 + NIP v3 (FirstApex) · 2026 Royal rebrand / digital-presence refresh.

N3.4 WS3 do-not-propose screening list

Screen out: any lending-use-case framing (out of scope here); any TVS Motor / Sundaram-Clayton / generic-TVS assumption; “build a distributor portal” (AMC has InvWise/Enhanced InWise; Alternates a distributor login; Royal iLink/iLounge/commercial portals); “introduce digital onboarding” (already present across all three); “use AI/ML in insurer claims” as if novel (Royal already claims ML/RPA/AI pilots); “modernise insurer core” without acknowledging FirstGen/NIP; any assertion that AMC/Alternates already have production-grade AI equivalent to Royal.

N3.5 WS3 full named-asset inventory

- **Sundaram AMC:** Investor App; InvWise App; Enhanced InWise; KFin-linked onboarding/KYC modification; MFU Online; BSE Star MF; NSE NMF; **MITRA**; riskometer-backed disclosures; Sundaram Business Cycle Fund; **GIFT City page.**
- **Sundaram Alternates:** SISOP, SELF, Voyager, Rising Stars, FIRST, SUNOPP, SUNIOP, SITP, Quant, India Equity, India Dedicated, Wonder Companies, Strategic Equity, Alpha Equity, ATLAS, ATLAS II, ACORN, Sundaram India Premier Fund, Performing Credit Opportunities Fund Series I, Emerging Corporate Credit Opportunities Fund I, RE Credit Fund IV; Wealth Connect login; digital onboarding; PMS-communication digitisation.
- **Royal Sundaram:** VConnect; iLink Agent Portal; iLounge; eInsure; **MOPIS**; commercial.royalsundaram portal stack; InsureX; IRCTC embedded travel-insurance issuance; FirstGen v7; NextGen Integration Platform v3; WhatsApp/SMS/email/app/web claims intake.

N3.6 WS3 room-role implications (the report’s own table)

AMC MD — balance growth, disclosure intensity, operating-control quality. AMC CBO — distributor economics, alliances, platform/service consistency across a very large distributor base. AMC senior fund manager — valuation/disclosure/riskometer/category discipline. AMC CFO — controls, fund-accounting ownership, expense transparency, resilience. AMC Chief Business Operations Officer — reconciliation, NAV cut-off control, branch+digital service consistency. Alternates MD — private-credit scaling, investor trust, closed-end discipline. Alternates fund manager — reporting credibility, lifecycle admin, conflict controls. Alternates zonal head — onboarding friction, partner enablement, investor-communication quality. Royal MD — separate growth ambition from operating reality across retail/commercial/embedded distribution. Royal COO — claims, servicing, resilience, grievance, cross-portal consistency. Royal CFO — underwriting discipline, commission/EOM governance, investment income. Royal CDO-Commercial — broker/intermediary cadence, commercial underwriting feedback, commercial-portal discipline. Royal CDO-Retail — channel activation, service continuity, proposition clarity (not channel invention).

N3.7 WS3 open questions / limitations

Exact current scheme count for Sundaram AMC; retail-vs-HNI-vs-institutional revenue/AUM mix; current PMS AUM split for Alternates; exact retail-vs-commercial premium split for Royal; named CDO-Commercial; exact public identity of in-room “senior fund manager”/“fund manager” attendees; entity-specific AI deployments at AMC and Alternates.

N3.8 WS3 evidence map

Question	Short answer	Confidence
AMC categories & scale	Equity/thematic/passive/hybrid/debt/liquid/FoF; avg MF AUM ₹64,746 cr; net AUM ₹63,217 cr	Verified
AMC client segments	Retail core; institutional/alliances present; HNI not split	Verified/Inferred
Alternates AIF/PMS & scale	Cat II & III; AIF ₹3,284 cr; platform ~₹7,000 cr+	Verified
Alternates dominant segment	HNI/UHNI/family-office skew	Verified/Inferred
Royal lines & retail/commercial split	Both broad; clean split not reconstructable	Verified/Inferred
Distribution & Royal dual structure	All channel-heavy; Royal separate retail/commercial portals; CDO-Commercial unnamed	Verified/Inferred
AMC operational pains	NAV/valuation, reconciliation, fund-accounting	Inferred

	integration	
Alternates operational pains	KYC/onboarding/accreditation, fund admin, reporting/conflict	Inferred
Insurer operational pains	Underwriting, claims orchestration, intermediary/portal	Verified/Inferred

N3.9 WS3 source register (cleaned)

Title	Publisher	Type	Confidence
AMC Annual Report 2024-25	Sundaram Mutual	Statutory AR	Verified
Mutual-fund schemes / product page	Sundaram Mutual	Official website	Verified
About Sundaram Mutual Fund	Sundaram Mutual	Official website	Verified
Management Team	Sundaram Mutual	Leadership page	Verified
Homepage / PMS / AIF product pages	Sundaram Alternates	Official website	Verified
PMS Disclosure Document (Mar 2024)	Sundaram Alternates	Regulatory disclosure	Verified
About Us	Sundaram Alternates	Official website	Verified
Annual Report 2024-25	Royal Sundaram	Statutory AR	Verified
About Us / Public Disclosures	Royal Sundaram	Official website	Verified
iLink / VConnect portal pages	Royal Sundaram	Official portal	Verified
FirstApex / rebrand / IRCTC-Insillion releases	Royal Sundaram	Official press release	Verified
Master Circulars: Mutual Funds / AIFs / Portfolio Managers; PM Regulations 2020	SEBI	Regulator	Verified
AIF & PMS statistics	SEBI	Regulator statistics	Verified
AMFI Monthly Note Mar 2025	AMFI	Industry body	Verified
Insurance Products Regs 2024; GI / Policyholders'-Protection master circulars; Maintenance-of-Information 2025; Cyber-Security Guidelines 2026	IRDAI	Regulator	Verified
Ramesh Babu CDO-Retail appointment	Trade press	Secondary	Inferred

N3.10 WS3 inference log

AMC pain concentrated in valuation/NAV/reconciliation — Medium; AMC retail-dominant not retail-only — High; Alternates dominant segment = affluent private capital — High; Royal public premium disclosures insufficient for clean retail/commercial split — High; Royal operates a real retail/commercial distribution split — High; Royal CDO-Commercial role exists operationally but publicly under-identified — Medium; AMC & Alternates have digital maturity without strong public AI claims — High.

N4 — Distribution & Aftermarket (WS4)

N4.1 Staged recommendations (facilitation prep)

- Stage 1 — before the immersion:** circulate the TSSFPL-vs-TVS-Mobility disambiguation one-pager with the OEM-franchise table; pull IMPAL’s latest AR and FY26 results to refresh branch/segment/B2B disclosure; request named org charts for Sundaram Motors (EVP-Cars, SGM-Finance) and MAS (EVP, CEO).
- Stage 2 — during:** prioritise MAS discovery (thinnest data); establish each entity’s real digital baseline before any opportunity framing; probe entity-specific pains and convert Inferred/Assumed tags to Verified or discard.
- Stage 3 — synthesis:** re-baseline against myTVS / Boodmo / Cars24 as family-adjacent comparators, not targets imposed on in-scope entities.

N4.2 WS4 source register (cleaned)

Title	Publisher	Type	Confidence
Corporate site & 70th Annual Report 2023-24	IMPAL	Company / filing	Verified
Ratings rationale	ICRA	Rating report	Verified
FY25 financials; MD-change filing	Screener/Groww/Tickertape	Market data / filing	Verified
TVS Group split structure	Business Standard	Press	Verified
TVS group restructured (branch ownership)	Industrial Economist	Analysis	Verified
Group corporate site	TSF Group	Group site	Verified
Mercedes-Benz Sundaram Motors; ED profile	Mercedes-Benz India	OEM/dealer site	Verified
Sundaram Honda (TVS Mobility) — out of scope	Sundaram Honda	Dealer site	Verified
MAS presence	Madras Auto Service (mastvs.com)	Company site	Verified (reliability flagged)
FY25 performance; aftermarket size	ACMA	Industry body	Verified
Aftermarket to ₹1,55,000 cr by FY30	ICRA via Businessworld	Industry/press	Verified
Replacement demand / EV impact (Dewan)	ICRA via Autocar Professional	Industry/press	Verified

Dealer profit pool (service & parts 49.6%)	NADA via Edmunds	Benchmark (US)	Verified
Counterfeit 30–40%	FICCI/ACMA via press	Industry/press	Verified (some dated)
Used-car market & GST; 5.41M used vs 4.16M new (2024)	CARS24 / Mordor / IMARC	Market reports	Verified
GoMechanic collapse & Spinny acquisition	Business Today / Tracxn / Inc42	Press	Verified
Boodmo	Boodmo / Entracker / MotorIndia	Company/press	Verified
myTVS digital-first aftermarket	MotorIndia / Tracxn	Press	Verified
Mercedes-Benz + Zoho SKYLine DMS	Motoring Trends	Press	Verified
TSERV Select / UTrust (TSSF-Toyota Kirloskar)	World Auto Forum / Autocar Pro	Press	Verified
DMS market in India	Research & Markets / AutoUnify	Market reports	Verified

(WS4’s evidence grid, function-by-function, 12 open questions and do-not-propose list are in M4.)

N5 — Shared Services & Social (WS5)

N5.1 Strategic-interpretation readiness stages

The three entities sit at distinct readiness stages: **SBS at “Built/Expert” (NASSCOM 3.10)** — isolated infrastructure, VLANs, ISO-governed delivery; **SMF at “Enthusiast” (2.50)** — centralised HIS/UHID, Digicare teams, ABDM-v3/FHIR transition; **Laxmi at “Explorer” (1.80)** — CMI tracking database, lightweight admin. Verified entity facts are separated from sector-wide inferences throughout (e.g. agentic SMSF processing, ambient scribing and predictive attrition are sector-derived *strategies*, not verified running implementations).

N5.2 WS5 source register (cleaned, representative)

Title	Publisher	Type	Confidence
“Looking to the Future” blog; Services / About / FAQ / Attrition-approach	Sundaram Business Services	Website	Verified
Departments / Outpatient / Overview / Academics / “Dear Akka”	Sundaram Medical Foundation	Website	Verified
Scholarship / Vridhi / mentorship pages	Laxmi Charities	Website	Verified
DPDP Act 2023 & Rules (Nov 2025)	MeitY / Gazette of India	Government	Verified
Revised Form CSR-1 notifications	MCA	Government	Verified
ABDM / SAHI / BODH strategy documents	Ministry of Health / NHA	Government	Verified
ICMR Ethical Guidelines for AI in Healthcare (2023)	ICMR	Government	Verified
AI Adoption Index	NASSCOM-EY	Sector research	Verified
APAC Front Line of Healthcare report (2026)	Bain & Company	Sector research	Verified
BPM / shared-services reviews	EY / Deloitte	Sector research	Verified

(WS5’s executive action map, milestone timeline, do-not-propose list, open questions and named assets are in M5.)

N6 — Group AI Posture (WS6)

N6.1 Full named AI/digital asset inventory

Asset / programme	Entity	Classification
IT Strategy Committee	Sundaram Finance	Governance
Cloud Data Lake; Systems of Intelligence layer	Sundaram Finance	Data platform / architecture
Machine-learning models; agentic-framework pilots	Sundaram Finance	AI / experimental AI
Intelligent Document Processing; 24x7 SOC	Sundaram Finance	Applied automation / cyber
FirstGen v7; NIP v3	Royal Sundaram	Core / integration platform
AI & digitisation pilots; ML + RPA (claims/ops)	Royal Sundaram	Experimental / applied AI
AMTIX; Vision Systems; cloud-server line traceability	Brakes India	Industrial automation / data layer
Investor App; InvWise	Sundaram Mutual	Digital channel / distributor platform

N6.2 WS6 source register (cleaned)

Title	Publisher	Type	Confidence
72nd Annual Report 2024-25; Familiarisation Programmes; Committee Composition	Sundaram Finance	AR / board docs	High
About SF / About TSF Group; Our Network	Sundaram Finance	Website	Medium-High
Annual Report 2024-25; FirstApex & 25-year releases	Royal Sundaram	AR / press	High

TSF Group home page	TSF Group	Group site	Medium-High
Home / AMTIX / Robotic Solutions pages; Sustainability Report 2023-24	Brakes India	Website / report	High
Home page	Sundaram Mutual	Website	High
Digital Lending Directions; IT Governance Directions	RBI	Regulator	High
Cyber-preparedness circular	IRDAI	Regulator	High
CSCRF	SEBI	Regulator	High
DPDP Rules / enforcement references	MeitY / India Code	Government	Medium-High

N6.3 WS6 inference log

Group AI ambition anchored more in lending than manufacturing — Inferred; manufacturing may be the best first thin-slice build surface — Inferred; no public group-wide AI CoE / central data office — Inferred; Brakes India CIDO likely spans enterprise IT + manufacturing digitisation — Inferred; public-disclosure inconsistency indicates fragmented information ownership — Inferred; responsible-AI capability lags cyber/privacy governance — Inferred. (WS6’s closing “facilitator takeaways” heading carries no content in the original.)

End of Part N. With Parts A–N, the consolidated pack now incorporates the full substantive content of all seven workstreams — including each report’s residual figures, strategic interpretations, open-questions and do-not-propose lists, named-asset inventories, milestone timelines, evidence maps and (cleaned) source registers. The only items not reproduced verbatim are the raw numbered URL citation lists, which remain in the original reports.